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AWN: A Community-Driven Donation and Volunteering Mobile Platform for Charities in Bahrain

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Abstract

This project is about AWN, an integrated donation and volunteering and campaign management platform for charities in Bahrain. AWN unites all services in a single mobile app for fundraising, giving charities, volunteers, and donors a seamless experience. Through this platform, users will be able to explore verified charities and their active campaigns, securely donate and register for volunteering, and view activities in real time. Charities have the tools to streamline management processes, update info and mobilize the community. Our app was created using FlutterFlow and powered by Supabase. As a result, it is secure, organized, and responsive. By using technology to connect and streamline the traditional giving process, AWN enhances transparency, increases efficiency, and brings communities together in working towards a shared cause – creating a new model for charitable giving in Bahrain.

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Chapter 1

Introduction

Supporting social welfare through charity work is essential, and yet many people in Bahrain don't have access to this wonderful opportunity. Often, donors are not aware of when these major donations are taking place and where these will have maximum effect. Just like that, people in need don't know which organization to get in touch with. Nor are they certain whether they can avail themselves of help for millions of other reasons. Moreover, volunteers often have trouble finding opportunities that suit their interests and time. These challenges indicate that there is no shared digital platform that links all the charitable parties in the Kingdom of Bahrain. To solve this problem, we launched an initiative to develop a brand-new application under the name AWN. It will incorporate all registered charitable associations in Bahrain under one framework. Through the platform, users will be able to search for associations, check their locations on Google maps, and participate in many charitable campaigns. The system aims to simplify all charitable processes for individuals and organizations by integrating donation, volunteering, and communication functions, thus increasing transparency, efficiency, and accessibility. The proposed platform will ensure that all charitable activities in Bahrain can take place through a single entity. This will enable donors, volunteers, and beneficiaries to communicate meaningfully so that they help reach the most deserving people. The charities will also promote their campaigns and manage resources and the public in a newly organized and impactful manner.

1.1 Problem Statement

Bahrain does not have an electronic system that covers all charities and their activities so far. Donors often don't know when and where the big fundraising campaigns are. Furthermore, they do not know where or how to donate in a timely manner. Beneficiaries have trouble identifying which organizations approach for help, and volunteers do not have clear access to community service opportunities. Users often find it difficult to look for the correct organization since they get a variety of inaccurate results while searching. When charity organizations are unregistered, it causes inefficiencies, duplication, and other lack of coordination issues. The AWN platform wants

to eliminate these problems by means of a centralized, transparent, and user-friendly system that links all the charitable players in the Bahrain ecosystem.

1.2 Project Objectives

The project aims to design and develop a single charity system for Bahrain which will bring together all the charities, donors, volunteers, and joint big campaign in one place. The app aims to make it easy to donate, recruit volunteers, and ask for help. They are also capable of hosting major campaigns which may be organized jointly by several charities and fully managed through the AWN app. Also, the project aims to assess the potential of this integrated system to enhance transparency, collaboration, and efficiency in the Charity Sector of Bahrain.

1.3 Relevance/Significance of the project

This work is crucial as it further strengthens the promotion of digital transformation in Bahrain's not-for-profit sector. By centralizing donation activity, it reduces selection of confusion, builds confidence, and capability for collaboration. The donation system yields ease, clarity, and transparency for the donors. It makes it easier for beneficiaries to access support and respond more quickly. It provides volunteers with opportunities to engage and participate in the community. First and foremost, they help charities to operate better, allow charities to stop duplication of effort and get further reaching the public. In short, AWN platform acts as a digital bridge that connects all players within the charity ecosystem in the Kingdom of Bahrain while creating stronger linkages of cooperation, transparency, and social responsibility.

1.4 Report Outline

This report is organized into several chapters that collectively present the motivation, development, and evaluation of the AWN Charity Application. Chapter 1 introduces the charitable landscape in Bahrain, outlines the problem, highlights the project objectives and significance, and explains the overall structure of the report. Chapter 2 provides a comprehensive literature review covering global digital donation systems, GCC charity applications, digital volunteering trends, and the technological factors influencing transparency and engagement, concluding with their relevance to the AWN platform. Chapter 3 discusses the project management approach, including the development methodology, timeline, risk analysis, mitigation strategies, and planned phases that guided the project's workflow. Chapter 4 describes the requirement collection process using surveys, observations, and document analysis, and it presents both functional and non-functional requirements along with personas, survey results, and system models. Chapter 5 focuses on the system design, showcasing the user interface, navigation flow, and structural design decisions that shape the AWN user experience. Chapter 6 explains the system implementation and testing procedures, detailing the use of FlutterFlow and Supabase, database structures, authentication mechanisms, administrative features, and the functional, usability, and performance tests conducted to ensure system reliability. Finally, Chapter 7 provides the project conclusion, summarizes key achievements, and suggests future enhancements to strengthen and expand the AWN platform.

Chapter 2

Literature Review

2.1 Overview of Charitable Organizations and Their Evolution

Giving to those in need has long been part of Islamic society, such as Bahrain, the Gulf States and Islam schism practices include zakat, which is mandatory, and Sadaqat, a voluntary charitable donation. Historically, donations were organized through mosques, local committees, or family networks that relied on personal trust. Volunteers or local leaders manually delivered donations, which were mostly in the form of cash or kind. This fostered community spirit but had low scale, traceability and accountability. As people moved to urban areas and more charities started up, donations and volunteer network management through in-person registrations, public announcements, or printed ads became unmanageable and fragmented. Despite this, trust and generosity continued to be present like earlier charitable systems (Abdelbaki, H.H. (2013)).

In the mid-20th century, the Gulf governments started to formalize charitable work through legal frameworks and registering societies under regulatory bodies such as Bahrain's Ministry of Social Development. During this period, the Royal Humanitarian Foundation, the Kuwait Red Crescent, Qatar Charity and other such charities was established. Through which these charities were refined into professional charities that follow structured programs, auditing and reporting standards. But efficiency and transparency were still being limited by manual processes, and that led to a slow move to digital. The inception of digital payment platforms like Bahrain BenefitPay was a transformative occurrence enabling secure, traceable, and accessible donations. Bahraini charities have improved donor confidence and increased accountability and operational reach with mobile payments and digital records, marking a shift from community to technology donation (Hunter, W. (2025)).

2.2 Digital Transformation of Charity Work

The digital revolution has changed how charities operate, communicate, and interact with the community. Nonprofit organizations are now able to manage donor information, financial records, and reporting processes more accurately and efficiently through the use of automation, cloud computing, and mobile technologies. Most smartphone use in Bahrain and the Gulf, at more than 95%, now drives mobile-first philanthropy. Online services allow donors to track their contribution in real time. Active participation is made possible by transparent auditing and an interactive dashboard displaying impact and contribution progress. Examples of initiatives where technology facilitates accountability, transparency, and participation are the Qatar Charity online hub, Saudi Red Crescent's volunteer portal, and Bahrain National eGovernment platform (von Liechtenstein, M. & Guillén, M. (2022)).

Mobile donations and digital volunteering systems are equally transformational as well. Digital donation services powered by mobile app payment solutions are now positioned to democratize giving. Applications such as Bahrain BenefitPay, UAE DubaiNow, Qatar charity app and Saudi Arabia Ihsan now offer secure, instantaneous donation features that provide donors with digital receipts. Such receipts enhance public trust and encourage repeat donations. In 2023 Ramadan, for example, more than 70% of contributions to Bahrain's Royal Humanitarian Foundation were made electronically, highlighting changing donor behavior. Additionally, these digital volunteer portals allow citizens to register, schedule, and verify community work online. Making civic service much more efficient and attractive particularly amongst the youth (Nahil Abdallah, A.,(2021)) While the benefits of these technologies improved efficiency, transparency and accessibility are obvious, there are still challenges concerning cybersecurity, digital literacy and trust among older donors. Therefore, charity digital transformation in the Gulf makes operations efficient and redefines donor culture. Thus, it strikes a balance between innovation and the traditional values of community support and compassion.

2.3 Global and Regional Perspectives on Digital Charity

2.3.1 Global Trends in Digital Donations

Worldwide, donations and other charitable giving have transformed dramatically and are done online nowadays. New studies reveal digital channels, especially online channels and mobile apps, now represent over 60% of global donations which is growing nearly three times faster than traditional giving (Empower Agency, 2025). Giving to charity is now a lot easier thanks to platforms including GoFundMe and JustGiving. Donors can now fund causes all over the world, without any geographical limitations as before. According to a study, live streams and influences cause a higher increase in participation. Furthermore, this factor can cause donation behavior to increase by 25%. Studies on the pandemic highlight the utility of business continuity (BC) and crisis management (CM) planning frameworks for effective strategy implementation. These frameworks help in strategizing responses to socio-economic shocks as well as in business malfeasance incidents. Unfortunately, not every place benefits from digital donations. Unlike North America and Europe where more than 70% of people donate in digital form, the Middle East and North Africa are still under 40% and that is mainly due to mistrust towards the platforms. Apart from this, it is also due to the failure to integrate local financial systems completely. (Bold.org Blog, 2025).

2.3.2 Digital Charity in the Gulf Cooperation Council (GCC)

Digital philanthropy is rapidly evolving across the GCC countries under national visions emphasizing social innovation and digital inclusion. Saudi Arabia's Ehsan platform, which launched in 2021, represents such progress, which has raised more than 1.5 billion SAR in its first year, integrates with 40+ registered charities, and incorporates national ID verification (Pearl Initiative, 2025). The "Smart Dubai" initiative in the UAE brought together all forms of donation on the DubaiNow app, which processes more than 60% of all donations, which provides a one-stop-shop for various other public services. While most donations taking place using mobile with more than 70% share, Qatar Charity has been using artificial intelligence to recommend campaigns to donors. The international extension of this model, along with transparency in Sharia compliance, is the e-donation portal for Direct Aid and Red Crescent.

These platforms leverage governance, technology, and cultural values to integrate zakat and Sadaqat within a digital structure that accounts for transparency and accessibility (Arab News 2025).

2.3.3 The Case of Bahrain within the Regional Context

Bahrain offers an innovative community approach to digital philanthropy. The Royal Humanitarian Foundation and Bahrain Red Crescent Society have created online donation portals that handled more than BD 1.2 million during the 2023 Ramadan, roughly 75% of the seasonal giving. Bahrain national e-wallet BenefitPay which collaborates with more than 100 licensed charities enabling cashless QR-based donation via mobile app, mosques and retail outlets is pivotal to their success. Bahrain has been supported by Tamkeen and the Information & eGovernment Authority and have prioritized non-profit digital readiness through grants, training and regulatory innovation. Bahrain can be an example for other small economies to adopt the same sustainable, tech-driven philanthropic model with the help of fintech, transparency, and cultural alignment. Older populations still face issues with digital hesitance and problems with cybersecurity which limits them. Bahrain steady progress demonstrates that uniformity and technological system can keep boosting donors’ trust in the charitable system.

2.3.4 Comparative Analysis of GCC Charity Applications

Table 2.3.4- 1. GCC Charity Applications

App / Platform	Country	Key Features	Unique Strength
Ehsan	Saudi Arabia	Nationally verified donation gateway, campaign transparency dashboards	Central government oversight ensuring authenticity
DubaiNow	UAE	Unified government app including donations, utility payments, fines	Seamless multi-service integration
Qatar Charity App	Qatar	AI-based personalized donation suggestions, multilingual UI	Advanced data analytics for donor engagement

Direct Aid	Kuwait	International relief focus, Sharia-compliant tracking	Cross-border project transparency
BenefitPay, Sadad	Bahrain	Instant mobile payments with integrated charity partners	National e-wallet linking banks and NGOs directly

The comparison highlights Bahrain distinctive approach of embedding charity giving within its national digital payment infrastructure, rather than developing a separate standalone donation app. This integration simplifies user behavior and can serve as the foundation for future apps.

2.4 Comparative Review of Existing Charity Applications

2.4.1 Overview of Global Charity Platforms

The global charity applications of GoFundMe, GlobalGiving and JustGiving have changed digital giving by combining convenience, transparency, and emotional storytelling. GoFundMe allows people to start their own fundraising campaigns where they can track their progress using updates and photos. The social media integration allows the campaigns to get just added visibility and donor reach (Magelssen, 2022). GlobalGiving serves as a trusted connection that verifies and links donors to organizations that are engaged in various social causes. Moreover, it brings authenticity and a measurable impact. JustGiving is a platform that links to social media and broadens the engagement and awareness of campaigns. These platforms collectively emphasize accessibility and storytelling, changing the way donors engage in philanthropy. There is an apparent reliance on international payment systems such as credit cards and PayPal. This may limit adoption in the Gulf region, where e-wallets and direct bank transfers are more common (SpringerOpen, 2024).

2.4.2 Charity Applications in the Gulf Region

Various charity applications introduced in the Gulf region show the importance of innovation, cultural circumstances, and regulatory oversight. In Saudi Arabia and Dubai, platforms are already enabling charitable giving with a range of payment options. Through predefined options, users can donate a preselected amount to any because they like. In a similar event, Qatar Charity’s mobile application uses AI (Artificial Intelligence) to recommend the most suitable campaign, alongside the Red Crescent e-Donation Portal from Kuwait which allows Sharia-compliant borderless charity

(Fiker Institute, 2024). The global world is slowly moving towards transparency and convenience of payments. However, what sets them apart is their robust linking to local e-wallet and identity verification systems. Resulting in a better trust level and accountability among donors (Badr Jafar, 2025).

2.4.3 Comparative Analysis of Features

Table 2.4.3- 2. Analysis of Features

Feature	Global Platforms (e.g., GoFundMe, GlobalGiving)	GCC Platforms (e.g., Ehsan, Qatar Charity, BenefitPay)
Verification	Varies by platform; some rely on campaign vetting only	Government or NGO-regulated to ensure Sharia compliance
Donation Methods	Credit cards, PayPal, and international banking	E-wallets, direct bank transfer, national payment systems
Focus	International crowdfunding and NGO fundraising	Local community aid, zakat, and relief projects
Transparency Tools	Campaign updates, reports, donor dashboards	Real-time tracking, verified organizations, official oversight
Volunteer Integration	Limited (mostly donation-based)	Often includes volunteering portals and training initiatives
Cultural Adaptation	Global audience	Faith-based and community-specific giving culture

This compares the global charity platforms that provide reach and personalization to GCC systems that provide trust, regulation, and faith knowledge. Bahrain's ecosystem is unique in that it blends fintech convenience with approved charities.

2.5 Volunteering and Community Engagement in the Digital Era

2.5.1 Evolution of Volunteering

Volunteering has been an essential part of community work for a long time. The local communities have participated in community development work, which is rooted in most Arab and Islamic culture, including sadaqah and social responsibility. In Bahrain and the wider Gulf region, early volunteering was facilitated by personal networks, mosque communities and/or word of mouth. Most activities were casual in nature where people willingly offered their time and effort for local events, relief campaigns, and Ramadan activities. As societies became more modern and populations expanded, these informal systems were not enough. It resulted in the formalization of voluntary actions through NGOs and state-affiliated institutions. By the late 2000s, Bahrain Royal Humanitarian Foundation and the Ministry of Labour and Social Development began registering volunteer groups, creating databases for participation, marking the beginning of recruitment (United Nations Volunteers, 2022).

2.5.2 Digital Transformation of Volunteering

Technology has totally changed the ways of volunteering in this century. Online platforms, now called ‘e-volunteering’ systems, allow people to register and find causes to volunteer remotely. These include tutoring, translation, and advocacy online to name a few. In Bahrain, the National Volunteer Platform was launched in 2021 under the Ministry of Youth and Sports Affairs of Bahrain. Thus, volunteers can register on the platform digitally. They can also select initiatives that fit their schedule, track their hours of participation, and receive official certificates of participation. This innovation helps in coordinating between organizations, creates accountability, and lowers operating costs by eliminating manual administrative work. Volunteers become even more flexible as well as convenient. This creates a positive participation factor with increased satisfaction levels that also enhances the culture of community service in the digital era (Lashin, 2022).

2.5.3 The Role of social media and Mobile Apps in Engagement

Today, social media plays a major role in mobilizing volunteers and raising awareness among the youth. Instagram, X (twitter), and WhatsApp help organize events, mobilize people, and inspire civic action. Youth groups in Bahrain are using these platforms to organize a clean-up on the beach, blood donation and Ramadan charity campaigns which is increasing participation. Reliance on digital communication shows that volunteering is not any more institution-led but embarking on a decentralization process. The Ehsan Volunteer Portal in Saudi Arabia and the Qatar Charity Volunteer App are other regional examples of how mobile technology is transforming volunteerism by providing citizens with attractive and accessible options to get involved in ongoing projects in the Gulf (Ahmed, 2022).

2.5.4 Challenges in Digital Volunteering

Digital volunteering has achieved notable success. Nonetheless, a few barriers persist that prevent them. Maintaining volunteer commitment and accountability is a major issue. Due to lack of follow up and unclear tasks, which online participants lack initial enthusiasm but fail to follow through. Moreover, older people and those from remote places may not have access to the right technology or digital literacy to take part appropriately. Cybersecurity and data privacy risks are challenging, as volunteer databases typically house sensitive personally identifiable information. Also, digital support can lack the emotional connections and sense of community that come with volunteering in person. In response to such challenges, numerous organizations in the Gulf region are utilizing hybrid models which incorporate online coordination, while also allowing for human interaction by maintaining some face-to-face volunteering (Alshammari, 2023).

2.5.5 Volunteering in Bahrain Charitable Ecosystem

Bahrain attempts to institutionalize volunteering reflect a greater awareness of its societal and developmental importance. With less than 20,000 registered volunteers currently in the Bahrain National Volunteer Platform and over 300 approved initiatives in the health, education and environmental fields (Ministry of Youth and Sports, 2025). Non-government organizations (NGOs) are also utilizing digital volunteers, such as the Royal Humanitarian Foundation and

Bahrain Red Crescent Society, to implement their humanitarian initiatives. This will enable citizens to register via mobile for service delivery, logistics, as well as donations.

2.6 Link to the Suggested AWN Application

According to the literature review, a gap clearly exists in Bahrain’s charitable ecosystem. Platforms like Saudi Arabia’s Ehsan, UAE’s DubaiNow, Qatar Charity, etc. have had great success in the digital transformation of philanthropy in GCC. Further, these systems either just cater to donation, are linked to government portals, or are missing integrates volunteering and local community options. Bahrain enjoys strong adoption of digital payments through BenefitPay. But has not come up with a unified system for donations, volunteers, beneficiaries, and charities. The suggested AWN application seeks to close this gap by providing a one-stop, integrated platform to facilitate secure digital donations, manage volunteers, track campaigns, discover charities, activate communities, and be assisted by AI. AWN incorporates features from GCC platforms and adds locally relevant features to create a more integrated and social experience relevant to Bahrain.

Table 2.6- 1. Comparison Between AWN and GCC Charity Platforms

Feature	AWN	GCC Platforms
Scope	Unified platform for donations, volunteering, charities & community	Mostly donation-only or government-service platforms
Volunteering	Fully integrated volunteer system	Rarely included (limited or none)
Charity Coverage	All registered Bahraini charities in one place	National charities, but fragmented across multiple apps
Community Features	Built-in community groups & engagement	Generally, not available
AI Support	AI assistant for guidance	Limited or not included
Transparency	Real-time campaign + volunteer tracking	Donation tracking only (campaign-based)
Payment Integration	BenefitPay + Apple Pay + cards	National e-wallets or government gateways

2.7 Summary

Various studies confirm that technology has changed the act of donation and volunteering from conventional donation through networking to faster, transparent, and data-driven systems. Due to digital platforms, charities are now able to improve participation, donor confidence, and cost. Bahrain Royal Humanitarian Foundation portal and BenefitPay offer a simple way to make online donations and pay digitally that involves the community.

Nevertheless, most of the existing systems remain fragmented and limited, fixed at monetary donations. Charities and volunteers are still not cooperating much. There is no holistic verified solution that connects donations, assistance requests and volunteer management. The AWN application proposal seeks to uptake filling this gap whereby charitable activity can take place in one singular environment which integrates everything together, thus allowing for better sharing of resources, transparency and inclusivity, all which help incorporate a culture of giving.

Chapter 3

Project Management

3.1 Process Model

The Awn Charity App project adopts an incremental development model with an Agile approach for flexibility, progress tracking, and continuous feedback. The given model aims to break down the system into small pieces that can be comprehended easily. Each of the increments delivers a working function like authentication, donation, and volunteer modules. Making use of Agile principles, development is done in short iterative cycles which allow continuous testing and enhancement after each stage. This hybrid approach allows requirements to change as per user and supervisor feedback and events while having a fixed schedule. It also reduces risks and helps detect errors early on. In addition, it ensures that every part of the project shows progress towards the final product.

3.2 Risk Management

Managing risks is important for the success of the Awn Charity App project. The goal is to identify anything that might disturb the scheduled delivery, quality, or performance of the project and plan on minimizing it. The risk management process will help the team stay prepared and maintain stability in the project's development phases since the project requires multiple technologies, data handling, and user interaction.

3.2.1 Risk Identification

The Awn project could experience a variety of risks during its life cycle. There can be technical issues here due to the Integration from Flutter Flow to Database or due to Android and iOS Risks related to schedules can occur if design and testing revisions take longer than anticipated. Resource risks can arise due to a lack of devices or tools needed for proper testing. There is also the issue of

data security, as the platform deals with sensitive donation information and user data. If the application interface or usability is lower than expected, user acceptance risks may arise.

3.2.2 Risk Analysis

The likelihood of each identified risk and the impact of its occurrence is assessed. For instance, there is a medium-to-high chance of technical and schedule risks occurring when the project goes into the design and integration phase. Whereas, the chance of a data security risk is low, however, it can have a very high impact. The analysis helps the team determine which risks to keep a closer eye on. If the team breaks down the risks into likelihood and consequence, they can focus efforts on the ones doing the most damage to the project.

3.2.3 Risk Mitigation Plan

The project team will take several measures to mitigate these risks. We will mitigate the technical risks by running tests on the Android and iOS versions early. Moreover, we will perform backups frequently. Using Flutter Flow debugging tools, we will further mitigate risk. We will manage schedule risks by providing a realistic Gantt chart and weekly reviews of our progress. The team will ensure that all required development tools are available, and emulators are ready to be tested for resource risks. The authentication and encryption features of the Database will be used for enhancing data security. To manage user acceptance risk, constant evaluation will be done through surveys and supervisor reviews to update the app's usability and design.

3.2.4 Risk Monitoring

We will keep a risk log throughout to ensure nothing crops up out of nowhere. The likelihood and mitigation actions of this risk will be recorded in the log. We will have frequent meetings with the supervisor for uncovering any new risk in time to mitigate it. The team consistently checks everything to ensure that no project objective is at risk. If there is some threat, they manage it before it develops.

3.3 Project activities Plan

All activities of the project were scheduled and organized in Microsoft Project for managing time, task, and dependence. The plan will run from June 1, 2025, to December 1, 2025, with Fridays and Saturdays as non-working days, but we sometimes work on these days. It has important steps called Initiation, Requirement & Analysis, Design, Development, Evaluation, and Testing & Reporting. Each of these phases has certain tasks and realistic durations. We drew the Gantt chart to visualize timelines and control progress. These activities must follow a logical order where some run concurrently, and other tasks depend on the completion of previous tasks. This helps the team to complete the project on time by doing everything smoothly.

The following table briefly presents the project activities with their durations and respective start and end dates. The sequencing of activities for the project was systematically carried out in order to ensure a realistic and orderly flow of work for all the phases in the project from initiation to reporting as illustrated below.

ID	Task Mode	Task Name	Duration	Start	Finish	Predecessors
0		AWN-CharityApp	115.5 days	Sun 6/1/25	Mon 12/1/25	
1		Initiation	19 days	Sun 6/1/25	Tue 7/1/25	
2		First meeting with supervisor	2 days	Sun 6/1/25	Mon 6/2/25	
3		Define project vision and scope	4 days	Tue 6/3/25	Mon 6/9/25	2
4		Conduct background research	5 days	Wed 6/4/25	Wed 6/11/25	
5		Identify project goals	4 days	Wed 6/11/25	Tue 6/17/25	3
6		Supervisor review and feedback	4 days	Thu 6/19/25	Wed 6/25/25	
7		Submit Porposal	3 days	Thu 6/26/25	Tue 7/1/25	6
8		Prepare project documentation and Gantt chart	5 days	Wed 6/11/25	Wed 6/18/25	
9		Requirement & Analysis	15.5 days	Wed 7/2/25	Thu 7/24/25	
10		Conduct surveys/interviews with users and charity organizations	6 days	Wed 7/2/25	Thu 7/10/25	
11		Define app features and system requirements	4 days	Sun 7/13/25	Wed 7/16/25	
12		Identify user roles (Donor, Volunteer, Admin)	3 days	Thu 7/17/25	Mon 7/21/25	11
13		Review requirements and obtain supervisor approval	3 days	Tue 7/22/25	Thu 7/24/25	12
14		Design	40 days	Sun 7/27/25	Thu 9/25/25	
15		Design mobile user interface using FlutterFlow	30 days	Sun 7/27/25	Tue 9/9/25	
16		Setup database and authentication system	7 days	Wed 9/10/25	Mon 9/22/25	15
17		Create ERD, DFD, and Use Case diagrams	4 days	Tue 8/12/25	Sun 8/17/25	
18		Review and adjust UI/UX design	1 day	Thu 9/25/25	Thu 9/25/25	17
19		Development	13 days	Sun 9/28/25	Fri 10/17/25	
20		Develop login and registration mo	3 days	Sun 9/28/25	Tue 9/30/25	
21		Implement donation and volunteer features	4 days	Wed 10/1/25	Tue 10/7/25	20
22		Add map integration and Whatsapp contact feature	3 days	Wed 10/8/25	Fri 10/10/25	21
23		Backend connection and function testing	3 days	Sun 10/12/25	Tue 10/14/25	
24		Optimize performance and finalize development	3 days	Wed 10/15/25	Fri 10/17/25	23
25		Evaluation Phase	9 days	Sun 10/19/25	Thu 10/30/25	
26		Conduct app testing on Android and iOS devices	2 days	Sun 10/19/25	Mon 10/20/25	
27		Fix bugs and improve usability based on feedback	4 days	Mon 10/20/25	Thu 10/23/25	
28		Evaluate system performance and collect results	2 days	Mon 10/27/25	Tue 10/28/25	27
29		Identify future improvements and feature ideas	1 day	Wed 10/29/25	Wed 10/29/25	28
30		Final evaluation review with super	1 day	Thu 10/30/25	Thu 10/30/25	29
31		Reporting & Finalization	25 days	Tue 10/21/25	Mon 12/1/25	
32		Prepare Report	20 days	Tue 10/21/25	Thu 11/20/25	
33		Prepare PPT	5 days	Sun 11/23/25	Thu 11/27/25	32
34		Final Review & Submit	0 days	Mon 12/1/25	Mon 12/1/25	33

Figure 3.3- 1. Project plan

The Gantt chart below depicts the schedule of the project. It shows the completion dates for all tasks. It will help us track the progress of the different activities in the various phases that will help us complete the project at the scheduled time.

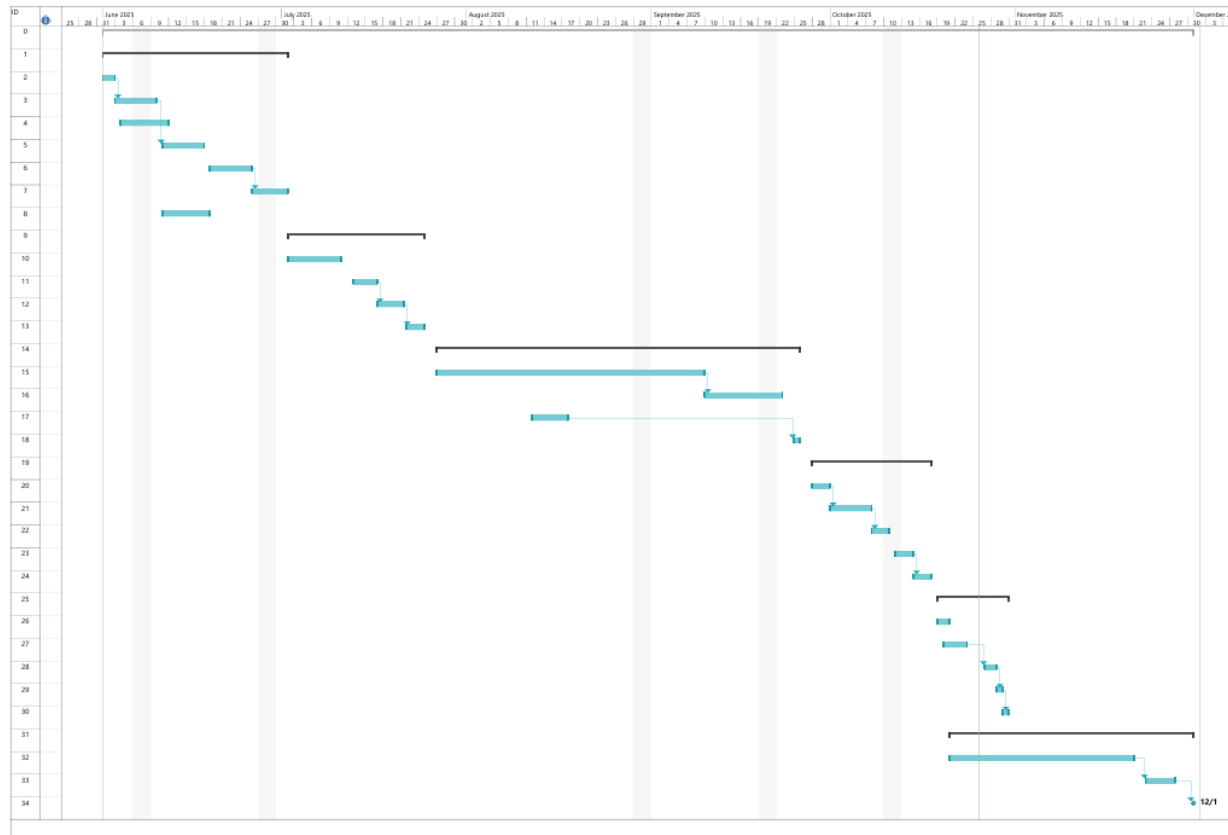


Figure 3.3- 2. Gantt Chart

Chapter 4

Requirement Collection and Analysis

Requirement gathering is an essential part of any system development process. It ensures that the final product meets user expectations and project goals. This chapter focuses on identifying, analyzing, and defining the functional and non-functional requirements of the AWN application. The AWN app aims to unify all charities, donors, and volunteers in Bahrain under one digital platform, enabling transparent, fast, and organized charitable participation.

4.1 Requirement Elicitation

The requirement elicitation process for the AWN application used both traditional and modern techniques to ensure a clear understanding of user needs. The main methods used were surveys, observation, and document review. An online survey was distributed to 188 participants with 16 questions, including donors, volunteers, and charity representatives, to gather feedback about their expectations and challenges when engaging in charitable activities. Additionally, existing charity platforms and donor platforms such as BenefitPay and the Royal Humanitarian Foundation portal were observed to identify their limitations and areas for improvement. Government charity regulations, volunteer registration policies, and charity website guidelines were also reviewed to ensure that the proposed system aligns with Bahraini standards. The collected information played a key role in defining user and system requirements that guided the design and implementation of the AWN application.

4.2 Survey Results

Below is a summary of the key questions and findings from the 188 responses:

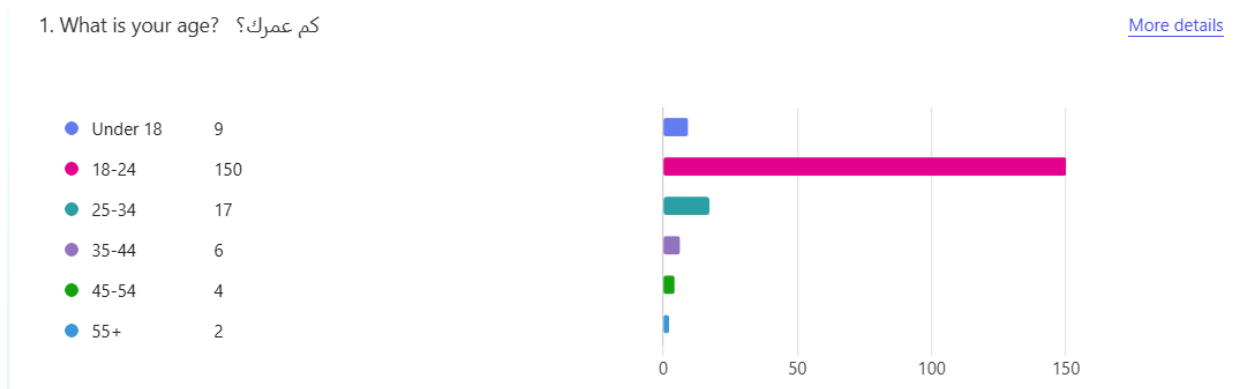


Figure 4.2- 1. Age Distribution of Respondents

As shown in Figure 4.2-1, Most respondents were aged 18–24, indicating that our survey primarily reached young adults who are highly engaged with digital platforms.

2. What is your gender? ما هو جنسك؟



Figure 4.2- 2. Age Distribution of Respondents

As shown in Figure 4.2-2, The responses were mostly from females (62%), showing higher engagement from women in participating in this survey.

3. Are you currently involved in charitable activities in Bahrain?

هل تشارك حاليًا في أنشطة خيرية في البحرين؟

● Yes, regularly نعم، بانتظام	17
● Sometimes أحيانًا	69
● Rarely نادرًا	51
● No لا	51

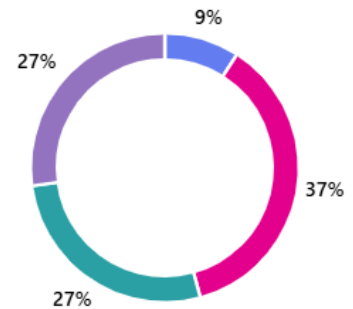


Figure 4.2- 3. Current Involvement in Charitable Activities

As shown in Figure 4.2-3, Most participants are involved only occasionally or rarely in charitable activities, indicating low consistent engagement.

4. How do you usually donate or volunteer? (select all that apply)

كيف تتبرع أو تتطوع عادةً؟ (اختر كل ما ينطبق)

● Directly with a charity office مباشرة عبر مكتب جمعية خيرية	54
● Bank transfer / cash تحويل بنكي / نقدًا	111
● Social media campaigns حملات على مواقع التواصل الاجتماعي	92
● Through friends or community groups عبر الأصدقاء أو المجموعات المجتمعية	61
● Other	7

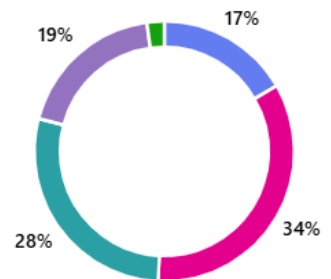


Figure 4.2- 4. Preferred Donation or Volunteering Methods

As shown in Figure 4.2-4, Most participants donate through bank transfers or social media campaigns, with fewer relying on direct charity visits or community groups.

5. What challenges have you faced when donating or volunteering?
ما هي التحديات التي واجهتك عند التبرع أو التطوع؟

● Lack of transparency (not knowing where the money goes) (انعدام الشفافية (عدم معرفة أين تذهب الأموال)	76
● Difficulty finding trustworthy charities (صعوبة العثور على جمعيات خيرية جديرة بالثقة)	21
● No easy digital way to donate or volunteer (لا توجد طريقة رقمية سهلة للتبرع أو التطوع)	38
● Limited awareness of available opportunities (الوعي المحدود بالفرص المتاحة)	69
● Other	56

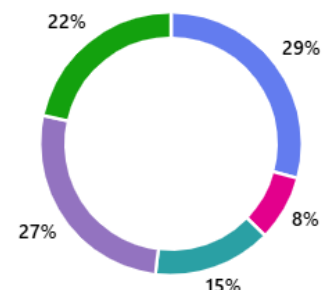


Figure 4.2- 5. Challenges Faced in Donating or Volunteering

As shown in Figure 4.2-5, Most respondents highlighted transparency issues and limited awareness of opportunities as the main challenges when donating or volunteering.

6. Would you prefer the platform as:
هل تفضل المنصة على النحو التالي:

● Mobile App only (تطبيق جوال فقط)	33
● Website only (موقع إلكتروني فقط)	34
● Both Them (كلاهما)	121

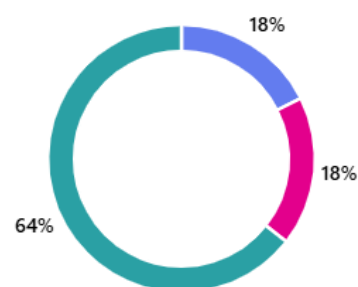


Figure 4.2- 6. Preferred Platform Type (App or Website)

As shown in Figure 4.2-6, Most participants prefer having both a mobile app and a website, showing a strong demand for multi-platform accessibility.

7. How important are the following features in a charitable platform?
ما مدى أهمية الميزات التالية في المنصة الخيرية؟

[M](#)



Figure 4.2- 7. Importance of Features in a Charitable Platform

As shown in Figure 4.2-7, Respondents rated all listed features as highly important, especially transparency, easy online donations, and connecting multiple charities in one unified platform.

8. How likely would you be to use a unified charitable app for Bahrain?
ما مدى احتمالية استخدامك لتطبيق خيري موحد في البحرين؟



Figure 4.2- 8. Likelihood of Using a Unified Charity App

As shown in Figure 4.2-8, A large majority of participants (70%) expressed strong willingness to use a unified charitable app, indicating clear demand for such a platform in Bahrain.

9. How often would you use a charitable app if available?
 كم مرة ستستخدم تطبيقًا خيريًا إذا كان متاحًا؟



Figure 4.2- 9. Expected Frequency of Using a Charity App

As shown in Figure 4.2-9, A significant portion of respondents (63%) would use a charitable app weekly or monthly, showing strong ongoing engagement potential for the platform.

10. How important is transparency in donations for you?
 ما مدى أهمية الشفافية في التبرعات بالنسبة لك؟



Figure 4.2- 10. Importance of Transparency in Donations

As shown in Figure 4.2-10, Most participants (84%) consider transparency highly important, confirming that clear tracking of donations is essential for trust.

11. What concerns would you have about using such a platform?
 ما هي المخاوف التي قد تكون لديك بشأن استخدام مثل هذه المنصة؟

● Security of my information أمن معلوماتي	51
● Misuse of donations سوء استخدام التبرعات	83
● Complicated or hard to use معقد أو صعب الاستخدام	31
● Lack of trusted charities قلة الجمعيات الخيرية الموثوقة	21
● Other	2

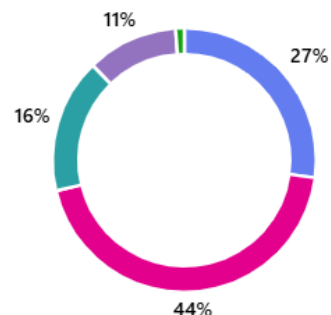


Figure 4.2- 11. Concerns About Using a Charitable Platform

As shown in Figure 4.2-11, Misuse of donations is the top concern among users (44%), highlighting the need for strict transparency and verified charities on the platform.

12. Have you volunteered before?
 هل سبق لك التطوع من قبل؟

● Yes	114
● No	74

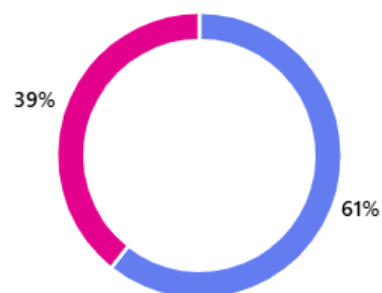


Figure 4.2- 12. Previous Volunteering Experience

As shown in Figure 4.2-12, Most respondents (61%) have previous volunteering experience, showing strong community engagement potential.

13. If yes, how often do you volunteer?
إذا كانت الإجابة بنعم، فكم مرة تتطوع؟

● Regularly بشكل منتظم	26
● Occasionally أحيانًا	66
● Rarely نادرًا	22

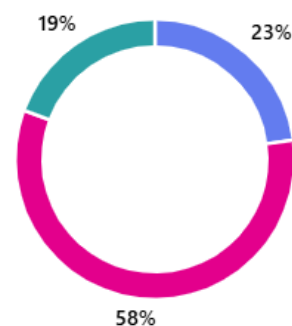


Figure 4.2- 13. Frequency of Volunteering Among Volunteers

As shown in Figure 4.2-13, Most volunteers participate occasionally (58%), indicating interest but limited consistency due to accessibility or opportunity gaps.

14. What challenges do you face in volunteering?
ما هي التحديات التي تواجهك في العمل التطوعي؟

● Lack of time ضيق الوقت	40
● Lack of information about opportunities قلة المعلومات عن الفرص المتاحة	63
● No easy way to register or track hours لا توجد طريقة سهلة للتسجيل أو تتبع الساعات	46
● Transportation or location issues مشاكل في المواصلات أو الموقع	20
● Other	2

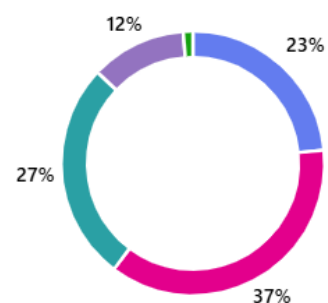


Figure 4.2- 14. Challenges Faced in Volunteering

As shown in Figure 4.2-14, Most respondents struggle due to limited information and unclear registration processes, making volunteering harder to access.

15. What would encourage you to volunteer more often?
 ما الذي قد يشجعك على التطوع أكثر؟

● Easy app sign-up for opportunities في سهولة التسجيل في التطبيق للفرص المتاحة	63
● Certificates or recognition شهادات أو تقدير	46
● Flexible volunteering schedules جداول تطوع مرنة	33
● Clear impact of my work تأثير واضح لعملي	44
● Other	2

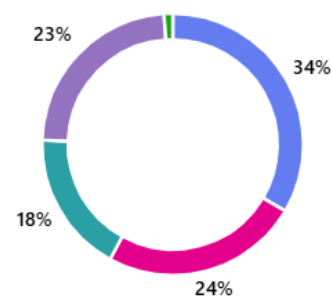


Figure 4.2- 15. Factors Encouraging More Volunteering

As shown in Figure 4.2-15, A simple sign-up process and clear recognition of volunteer efforts are the strongest motivators for increasing participation.

16. Do you have any suggestion before start design app (if no you can skip) هل لديك أي اقتراح قبل البدء في تصميم التطبيق (إذا لم يكن لديك يمكنك التخطي)

36 Responses

ID ↑	Name	Responses
1	anonymous	.
2	anonymous	Prioritize Ease of Use, Security of Transactions and User Information and Data, and App Efficiency (Not Slow)
3	anonymous	No , go ahead and design
4	anonymous	يكون باللغتين العربي والإنجليزي
5	anonymous	The app should support English and Arabic language
6	anonymous	لا يوجد
7	anonymous	-
8	anonymous	اقترح اخذ فكرة تطبيق قطر الخيرية وهي سحب مبلغ يومي من الحساب يحدده المستخدم من دون الحاجة لدخوله وتبرعه بنفسه
9	anonymous	Categories for each charity case

Figure 4.2- 16. Respondents' Suggestions Before Designing the App

As shown in Figure 4.2-16, Most suggestions emphasized ease of use, strong security, bilingual support, clear categorization for charities, and ensuring the app runs efficiently without performance issues.

4.3 System Requirements

This section outlines the functional and non-functional requirements for the AWN system. Functional requirements define the system behavior and user interactions, while non-functional requirements focus on quality attributes such as security, performance, and usability.

Functional Requirements:

Table 4.3-1: FRQ01 – User Registration

Requirement ID	FRQ01
Requirement Name	User Registration
Requirement Description	Allow users (donors, volunteers, or charities) to register by entering personal details such as name, email, password, and user type.
Priority	High
Reason	Enables access to system features and personalized experience.
Measurement	Number of successful registrations.
Requirement Status	Proposed

Table 4.3-2: FRQ02 – Login/Logout

Requirement ID	FRQ02
Requirement Name	Login/Logout
Requirement Description	Enable users and administrators to log in securely using credentials and log out safely after use.
Priority	High
Reason	Ensures secure and private access to user accounts.
Measurement	Successful login/logout rate.

Requirement Status	Proposed
---------------------------	----------

Table 4.3-3: FRQ03 – Browse Charities

Requirement ID	FRQ03
Requirement Name	Browse Charities
Requirement Description	Users can browse all registered charities, view descriptions, phone numbers, locations, and active campaigns.
Priority	High
Reason	Helps users easily find and connect with charities across Bahrain.
Measurement	Number of charities listed and accessed.
Requirement Status	

Table 4.3-4: FRQ04 – Make Donations

Requirement ID	FRQ04
Requirement Name	Make Donations
Requirement Description	Allow users to make secure online donations through BenefitPay, bank transfer, or credit card.
Priority	High
Reason	Core function enabling digital giving and transparency.
Measurement	Total number and amount of donations.
Requirement Status	Proposed

Table 4.3-5: FRQ05 – Volunteer Registration

Requirement ID	FRQ05
Requirement Name	Volunteer Registration
Requirement Description	Enable users to register as volunteers and apply for campaigns or charity events.
Priority	Medium
Reason	Encourages organized and traceable volunteer participation.
Measurement	Number of volunteer sign-ups.
Requirement Status	Proposed

Table 4.3-6: FRQ06 – Campaign Management (Admin)

Requirement ID	FRQ06
Requirement Name	Campaign Management (Admin)
Requirement Description	Allow administrators to add, update, or delete campaigns including titles, goals, and duration.
Priority	High
Reason	Keeps campaigns accurate and up to date.
Measurement	Number of active campaigns managed.
Requirement Status	Proposed

Table 4.3-7: FRQ07 – Charity Management (Admin)

Requirement ID	FRQ07
Requirement Name	Charity Management (Admin)
Requirement Description	Allow administrators to insert new charities, edit details (location, contact, description), or remove inactive ones.
Priority	High
Reason	Ensures reliable and accurate charity information in the database.
Measurement	Number of charity records updated.
Requirement Status	Proposed

Table 4.3-8: FRQ08 – Volunteer Management (Admin)

Requirement ID	FRQ08
Requirement Name	Volunteer Management (Admin)
Requirement Description	Enable admin users to manage volunteer profiles by adding, editing, or deleting volunteer records.
Priority	Medium
Reason	Maintains an active and verified volunteer base.
Measurement	Number of volunteer records maintained.
Requirement Status	Proposed

Table 4.3-9: FRQ09 – Progress Tracking

Requirement ID	FRQ09
Requirement Name	Progress Tracking
Requirement Description	Display real-time progress bars for campaigns, showing funds collected and volunteer participation.
Priority	High
Reason	Promotes transparency and motivates user engagement.
Measurement	Accuracy and update frequency of progress data.
Requirement Status	Proposed

Table 4.3-10: FRQ10 – Notification System

Requirement ID	FRQ10
Requirement Name	Notification System
Requirement Description	Send real-time alerts to users about donation confirmations, campaign updates, and volunteer approvals.
Priority	Medium
Reason	Keeps users informed about their activity and new opportunities.
Measurement	Notification delivery success rate.
Requirement Status	Proposed

Non-Functional Requirements:

Table 4.3-11: NFRQ01 – Security

Requirement ID	NFRQ01
Requirement Name	Security
Requirement Description	Protect all data with encryption, secure authentication, and password hashing (SSL/TLS).
Priority	High
Reason	Ensures user confidentiality and system integrity.
Measurement	Compliance with SSL/TLS standards.
Requirement Status	Proposed

Table 4.3-12: NFRQ02 – Performance

Requirement ID	NFRQ02
Requirement Name	Performance
Requirement Description	System should handle concurrent users efficiently and respond to requests in under 2 seconds.
Priority	High
Reason	Ensures smooth performance for all users.
Measurement	Response time monitoring.
Requirement Status	Proposed

Table 4.3-13: NFRQ03 – Usability

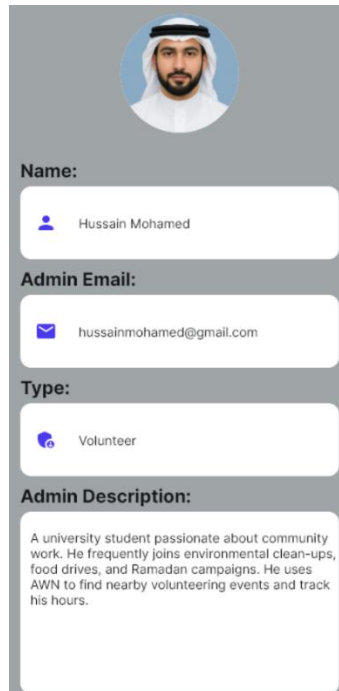
Requirement ID	NFRQ03
Requirement Name	Usability
Requirement Description	App should be easy to navigate, with simple icons and instructions for all age groups.
Priority	Medium
Reason	Enhances accessibility for users of varying digital literacy.
Measurement	User satisfaction survey results.
Requirement Status	Proposed

Table 4.3-14: NFRQ04 – Availability

Requirement ID	NFRQ04
Requirement Name	Availability
Requirement Description	The application should be available 24/7 with minimum downtime.
Priority	High
Reason	Keeps donation and volunteering services always accessible.
Measurement	System uptime percentage.
Requirement Status	Proposed

4.4 Personas

Four personas were defined to better understand the different user groups that interact with AWN. These would cover the main roles within the system, i.e., volunteer, donor, charity rep, and admin. They help the ensure that the end-user need is in line with the design of the system.



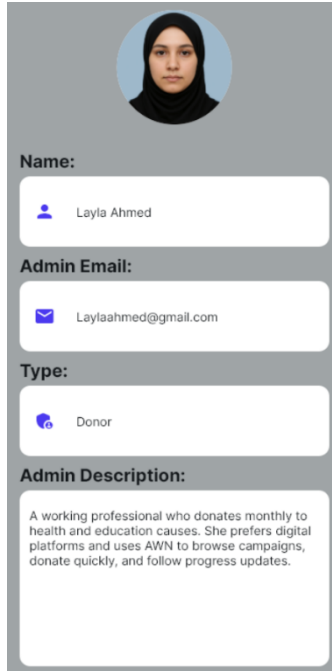
A vertical form representing a user profile. At the top is a circular profile picture of a man with a beard wearing a white thobe and ghutra. Below the picture are four sections, each with a label and a text input field:

- Name:** Hussain Mohamed
- Admin Email:** hussainmohamed@gmail.com
- Type:** Volunteer
- Admin Description:** A university student passionate about community work. He frequently joins environmental clean-ups, food drives, and Ramadan campaigns. He uses AWN to find nearby volunteering events and track his hours.

Figure 4.4 -1. Persona Profile: Hussain Mohamed

Persona 1: Volunteer

Hussain is a representation of young Bahrainis who volunteer in activities like beach clean-ups, food drives, and Ramadan charitable events. He uses AWN to find opportunities near him, register immediately, and track hours online. He wants to help society efficiently and also keep a clear service record. Hussain appreciates simplicity, events that are easily accessible, and timely notifications to stay in touch.



A digital persona profile card for Layla Ahmed. At the top is a circular profile picture of a woman wearing a black hijab. Below the picture, the card is divided into sections with labels and input fields. The 'Name' section shows 'Layla Ahmed' with a person icon. The 'Admin Email' section shows 'Laylaahmed@gmail.com' with an email icon. The 'Type' section shows 'Donor' with a heart icon. The 'Admin Description' section contains a paragraph of text.

Name:

Layla Ahmed

Admin Email:

Laylaahmed@gmail.com

Type:

Donor

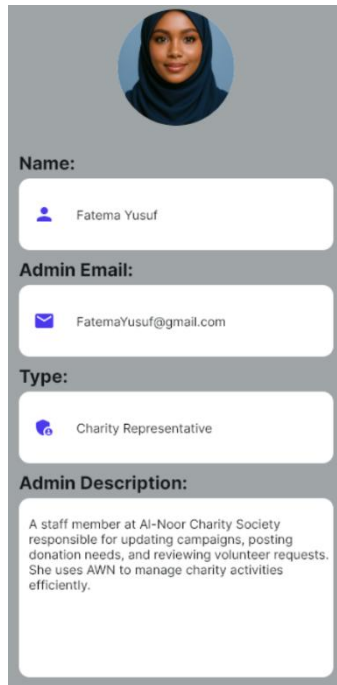
Admin Description:

A working professional who donates monthly to health and education causes. She prefers digital platforms and uses AWN to browse campaigns, donate quickly, and follow progress updates.

Figure 4.4 -2. Persona Profile: Layla Ahmed

Persona 2: Donor

Layla represents modern donors who prefer fast, digital giving rather than traditional cash donations. As a working professional, she uses AWN to browse active campaigns, donate securely, and follow transparent progress indicators. She appreciates being able to view charity information, verify legitimacy, and receive digital receipts instantly. Layla's primary motivation is contributing reliably to causes related to health, education, and emergency relief without facing the confusion of scattered platforms.



A vertical user profile card with a grey background. At the top is a circular profile picture of a woman wearing a blue hijab. Below the picture, the card contains four white input fields, each with a label and an icon. The first field is labeled 'Name:' with a person icon and contains the text 'Fatema Yusuf'. The second field is labeled 'Admin Email:' with an envelope icon and contains the text 'FatemaYusuf@gmail.com'. The third field is labeled 'Type:' with a group of people icon and contains the text 'Charity Representative'. The fourth field is labeled 'Admin Description:' and contains a paragraph of text.

Name:

Fatema Yusuf

Admin Email:

FatemaYusuf@gmail.com

Type:

Charity Representative

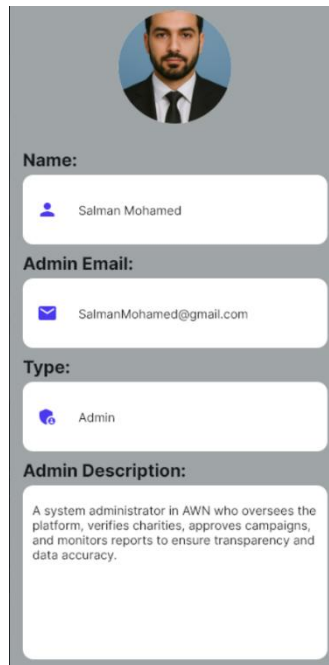
Admin Description:

A staff member at Al-Noor Charity Society responsible for updating campaigns, posting donation needs, and reviewing volunteer requests. She uses AWN to manage charity activities efficiently.

Figure 4.4 -3. Persona Profile: Fatema Yusuf

Persona 3: Charity Representative

Fatema manages activities at Al-Noor Charity Society. She updates campaigns, posts donation needs, reviews volunteer requests, and engages with the community using AWN. Her day-to-day tasks include talking to the charity staff and ensuring that the campaigns have the correct information. Fatema wants to make charity work easier by using digital technology so they cut down on paperwork and can reach more donors and volunteers all through one platform. AWN helps her manage all activities from one dashboard.



A vertical user profile card with a grey background. At the top is a circular profile picture of a man with a beard and a suit. Below the picture are four white input fields, each with a grey header and a light blue icon on the left. The first field is for the name 'Salman Mohamed' with a person icon. The second is for the admin email 'SalmanMohamed@gmail.com' with an envelope icon. The third is for the type 'Admin' with a key icon. The fourth is for the admin description, containing a paragraph of text.

Name:

Salman Mohamed

Admin Email:

SalmanMohamed@gmail.com

Type:

Admin

Admin Description:

A system administrator in AWN who oversees the platform, verifies charities, approves campaigns, and monitors reports to ensure transparency and data accuracy.

Figure 4.4 -3. Persona Profile: Salman Mohamed

Persona 4: System Administrator

Salman is a platform administrator working in AWN which involves running the program, data accuracy, and compliance. He verifies charity accounts and approves campaigns. He also monitors the system's activities and handles transparency and safety reports. He ensures that donations, volunteer records and campaign information comply with Bahrain regulations and digital security. Salman aims to create a safe and effective environment for charities, donors and volunteers to interact with each other.

4.5 System Models

The system diagrams of the AWN platform provide an overall view of how the data moves through the application, how the user actions are processed and how the system components work together. This includes Data flow diagrams DFDs of level 0 and level for users and admin. The key processes in this are registration, authentication, charity, and campaign browsing, donation, volunteer, and communication management. Process specifications break down each major function into its inputs and outputs, internal logic and behavior. Also, the ERDs explain how data should be stored in the database. Furthermore, it shows the relationship between the users, donation, volunteers, campaign, charity, notifications, and administration. To ensure a correct understanding of system functionalities, the accurate development of software, and a well-structured approach for implementing and validating the AWN application.

4.5.1 Data Flow Diagram

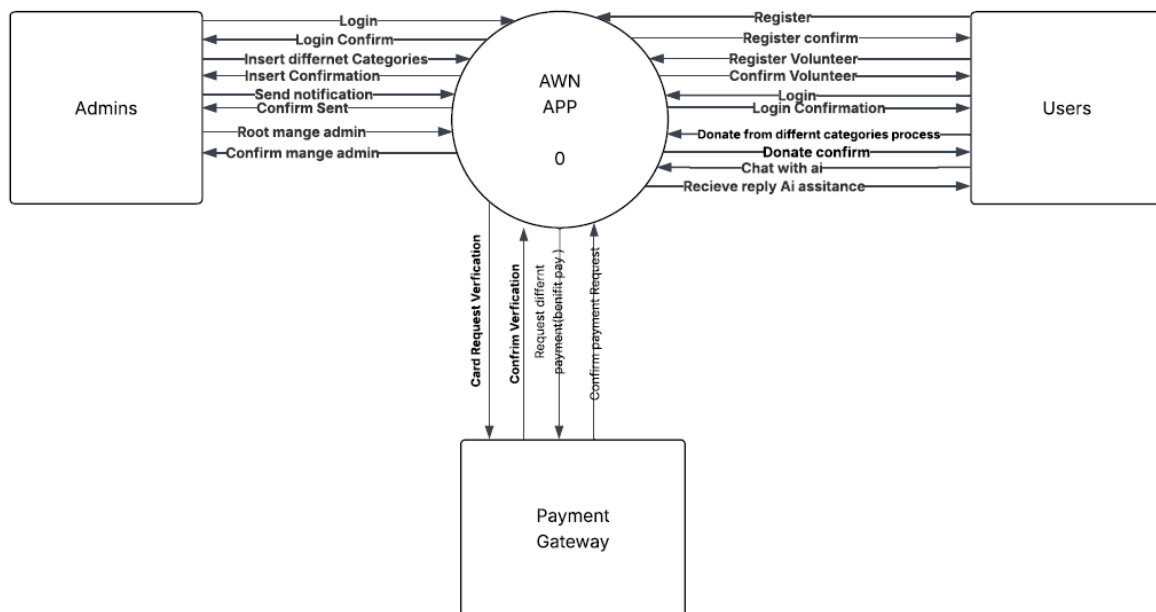


Figure 4.5.1 -1. Data Flow Diagram Level 0

The diagram represents the Level 0 Data Flow Diagram (DFD) of the AWN platform, which illustrates the connection and data interchange between the Users, Admin, Payment Gateway and the system. Users will interact with the system through registration, login, donations, volunteering request, charities and campaign discovery and communication with the help of a chatbot as shown in Figure 4.5.1-1. The system provides notifications, confirmations as well as access to available feature and services in return. The platform administrator manages who can login to the system and updates the charity or campaign info. He reviews and approves requests, send notices, and what the system contains at a general level.

The system securely processes donation transactions through a payment gateway and responds with confirmations. To conclude, AWN allows users and administrators to connect to the payment gateway in a functional manner by using one centralized system, as shown in the Level 0 diagram.

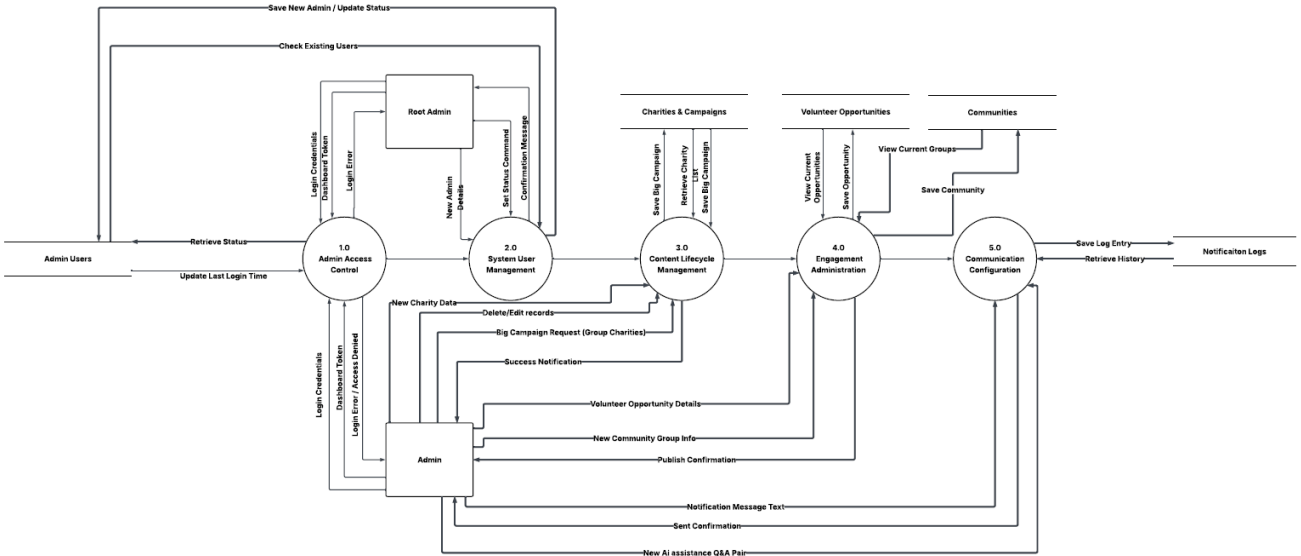


Figure 4.5.1 -2. Data Flow Diagram Level 1- Admin Module

This is the DFD Level 1 for the AWN Admin Module, as shown in Figure 4.5.1- 2. The admin operations are defined into five processes. These processes are Admin Access Control, System User Management, Content Lifecycle Management, Engagement Administration, Communication Configuration. The process starts with the login verification for secure access to the administrator. After a successful login, the admin can manage user accounts, edit permissions, or deactivate users. Content Lifecycle Management is responsible for the update and approval of charity details and campaigns. Engagement Administration is responsible for the activation of volunteer opportunities and events in your community. Lastly, Communication Configuration deals with updates notifications and message logs for chatbots. The connected processes of an automated work notification ensure that the administrative tasks of AWN are regulated and organized. Further, this ensures a streamlined and efficient operation of the system.

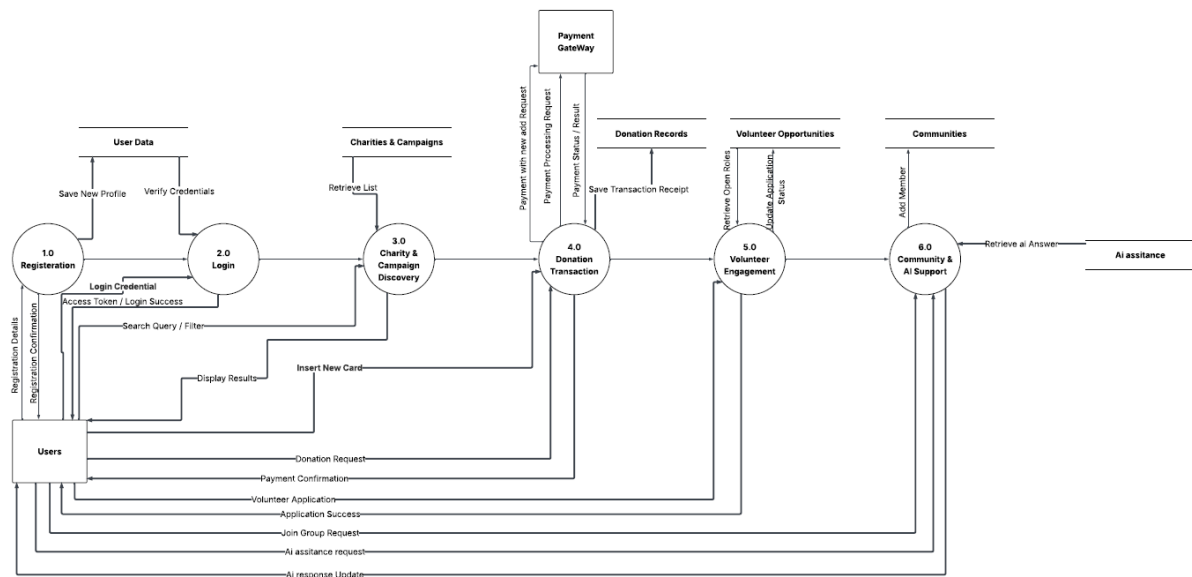


Figure 4.5.1 -3. Data Flow Diagram Level 1- Users Module

As illustrated in figure 4.5.1-3, the DFD Level 1 for user module describes user interaction with the AWN through six processes, including Registration, Login, Explore Charities and Campaigns, Donation Transaction, Volunteering, and Community & AI. The interaction starts with the user creating an account and logging in. The system validates the details and securely logs the user in. After login, user will be able to browse through charities/campaigns, search as per interest, make secure donation through integrated payment gateway, and track confirmed donation record. People can also browse, apply and receive updates related to volunteering opportunities through their profile. Through the platform, people can easily engage with the community and connect with AI for information or guidance. All records of users, donations, volunteers and chatbots get updated from all places continuously with full integration.

4.5.2 Process Specification

Level 1 for AWN (for Admin):

The admin processes of the site are reported in these pages. All Level-1 processes are expanded with process specification which states its purpose, inputs, and output. They are: Admin authentication control, system user management, content lifecycle management, engagement administration and communication configuration. The tables in this section provide additional clarity on how each are processes works and how data flows in the system from an administrative perspective.

Table 4.5.2- 1: DFD Level 1 – Admin Access Control

Process ID	1.0
Process Name	Admin Access Control
Process Description	Handles admin authentication, verifies login credentials, checks account status, and grants or denies system access.
Process Input	Login Credentials (Username, Password) Authentication Token Request
Process Output	Successful Login Confirmation Access Token Issued Login Error Updated Last Login Record

Table 4.5.2- 2: DFD Level 1 – System User Management

Process ID	2.0
Process Name	System User Management
Process Description	Manages admin accounts by creating, updating, activating, or disabling admin users, including role assignment and access rights.
Process Input	New Admin Details Update/Delete Command
Process Output	Status Confirmation Message Updated Admin Database

Table 4.5.2- 3: DFD Level 1 – Content Lifecycle Management

Process ID	3.0
Process Name	Content Lifecycle Management
Process Description	Handles creating, updating, retrieving, and deleting charity data, campaigns, and volunteer opportunities, ensuring information is always accurate and up to date.
Process Input	New Charity or Campaign Data Edit/Delete Command Retrieval Request
Process Output	Updated Content Record Success/Failure Notification

Table 4.5.2- 4: DFD Level 1 – Engagement Administration

Process ID	4.0
Process Name	Engagement Administration
Process Description	Manages volunteer opportunities, community groups, and engagement content. This process ensures creation, updating, and retrieval of engagement activities within the platform.
Process Input	Volunteer Opportunity Details Community Group Information Publish / Edit Requests
Process Output	Published Opportunities / Groups Status or Confirmation Message

Table 4.5.2- 5: DFD Level 1 – Communication Configuration

Process ID	5.0
Process Name	Communication Configuration
Process Description	Allows the admin to manage communication settings, notifications, and chatbot knowledge updates within the platform.
Process Input	Notification Message Text New Chatbot Q&A Pair Log Entry Data
Process Output	Updated Communication Rules Saved Log Entries Status or Confirmation Message

Level 1 for AWN (for User):

In this section, the major user functions from the Level 0 diagram are broken down into more detailed sub-processes that show how the user interacts with the system. The registration, login, browsing charities and campaigns, donating, volunteering and getting support services is broken down to show how data is passed between the user, the various components of the system, as well as external systems such as the payment gateway. The tables given below show the Levels 1 process with their inputs, outputs and brief description, it gives you a clear idea about the behaviors of the system from the user point of view.

Table 4.5.2- 1: DFD Level 1 – Registration

Process ID	1.0
Process Name	Registration
Process Description	This process enables users to create an account by submitting their details, which the system verifies and securely stores for future access.
Process Input	Registration Details (Name, Phone, Email, Password) Verification Request
Process Output	Registration Confirmation Stored User Profile

Table 4.5.2- 2: DFD Level 1 – Login

Process ID	2.0
Process Name	Login
Process Description	This process verifies the user's login credentials and grants access if authentication is successful.
Process Input	Login Credentials (Email/Phone & Password)
Process Output	Login Confirmation / Access Token Error Message if authentication fails

Table 4.5.2- 3: DFD Level 1 –Charity & Campaign Discovery

Process ID	3.0
Process Name	Charity & Campaign Discovery
Process Description	This process retrieves and displays available charities and campaigns so users can browse and select based on their interests.
Process Input	Search Query / Filters
Process Output	Displayed Charity & Campaign Results

Table 4.5.2- 4: DFD Level 1 –Donation Transaction

Process ID	4.0
Process Name	Donation Transaction
Process Description	This process securely handles donation requests by sending payment details to the payment gateway and returning a success or failure response to the user.
Process Input	Donation Request Payment Confirmation
Process Output	Donation Receipt / Status Message

Table 4.5.2- 5: DFD Level 1 –Volunteer Engagement

Process ID	5.0
Process Name	Volunteer Engagement
Process Description	This process allows users to view available volunteer opportunities and submit applications, updating their status throughout the approval process.
Process Input	Volunteer Application Update Application Status
Process Output	Application Success Message Volunteer Status Notification

Table 4.5.2- 6: DFD Level 1 –Community & Support Hub

Process ID	6.0
Process Name	Community & Support Hub
Process Description	This process allows users to access community content, receive support, and interact with FAQs, helpdesk, or updates created by the admin.
Process Input	Request Support View Community Section
Process Output	Help and Support Response Community Content Display

4.5.3 Entity Relationship Diagram

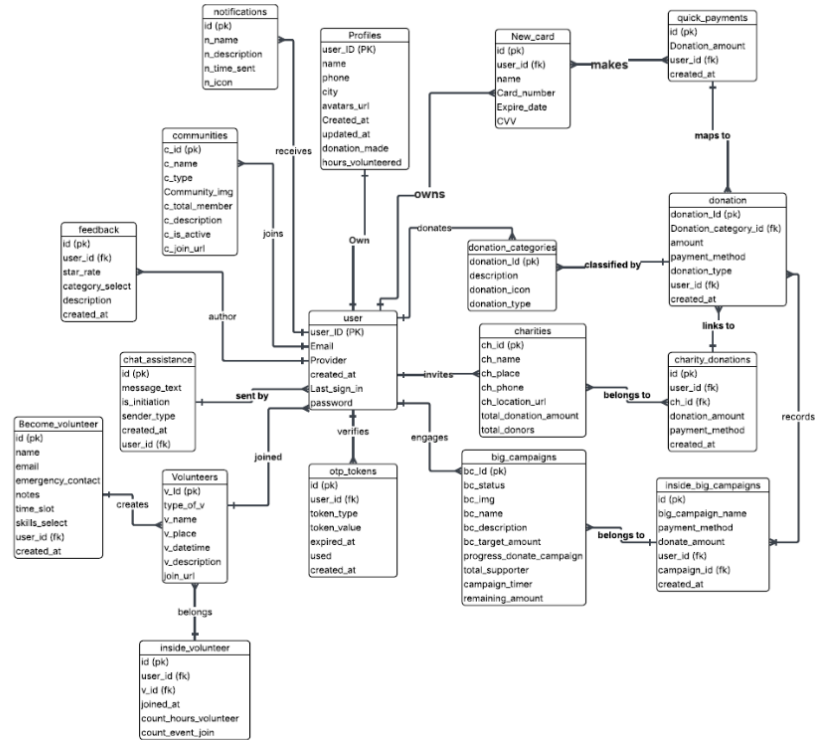


Figure 4.5.3- 1. Entity Relationship Diagram for user

According to Figure 4.5.3-1, this ERD shows how users participate in donation, volunteering, campaigning and community features in the AWN. The Profiles table in the model which is the central device in the database – it holds important user data as well as connects to several other systems. Entities related to donation, such as Donations, Charity_Donations, Big_Campaigns, and Quick_Payments, contain payment details, donation history, and campaign details. The process of volunteering is supported by Become_Volunteer, Volunteers and Inside_Volunteer, which can track the submission, approval, attendance and number of volunteer hours. Communities and Feedback, which convey community engagement, enables users to join communities, and provide structured reviews and comments. The system provides notifications, chat assistance and chatbot agent for communication. OTP Tokens help in securing user access. The diagram shows how every activity of the user such as donating, volunteering, communicating, community-engagement relate to the system which is integrated and secure.

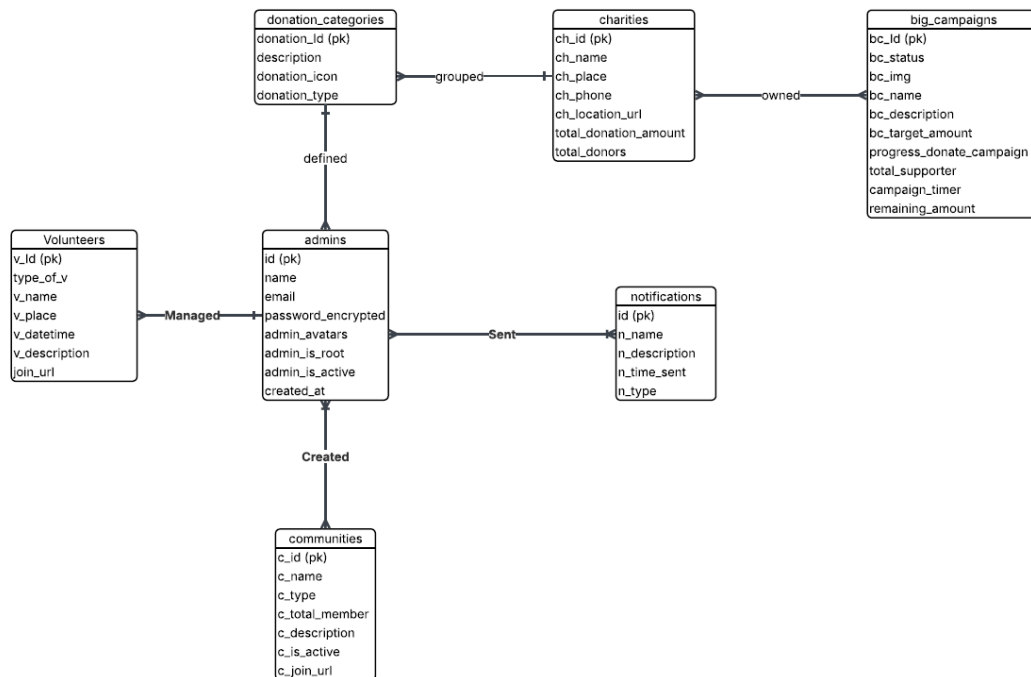


Figure 4.5.3- 2. Entity Relationship Diagram for Admin

As shown in Figure 4.5.3-2, administrators can manage and control system functions through this ERD from AWN. The Admins table is the main table used to store users' credentials, permissions, and access. With the help of this structure, the administrator handles charity records, donation categories and other system content with the help of related tables such as Charities, Big_Campaigns, Inside_Big_Campaigns and more. Also, the administrator can manage volunteer management with the help of this ERD. The volunteers and become volunteer will help to approve the volunteers and create roles for them. Admin can use Notifications to keep users informed with updates, alerts and messages from their system. In general, the diagram highlights that the administrative environment manages content, manages operations and controls the system to ensure system is operated, secure and orderly.

Chapter 5

System Design

5.1 Introduction

This chapter provides a comprehensive overview of the system design for the AWN Charity Application, illustrating how the platform's functional and non-functional requirements were translated into a structured, user-friendly, and visually coherent interface. It describes the architectural decisions behind the layout, navigation flow, and screen organization, ensuring that each component supports efficient interaction for donors, volunteers, charities, and administrators. The chapter presents detailed design representations of key features such as authentication, donation processing, volunteer registration, campaign browsing, community engagement, and administrative control panels, highlighting how usability and clarity were prioritized throughout the design process. By showcasing the arrangement of UI elements, visual hierarchies, and the logic behind each page, this chapter establishes a clear blueprint that guided implementation and ensures that the system delivers a seamless charitable experience aligned with the project's goals.

5.2 Splash Screen:

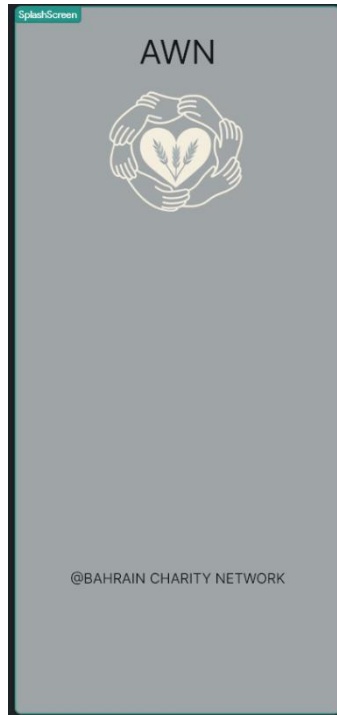


Figure 5.2- 1. Splash Screen

The splash screen shows the logo of the AWN application as well as the name of the application as well as the identity of the development team, as shown in Figure 5.2-1. This is the first view of the platform that creates a brand identity while the system is loading.

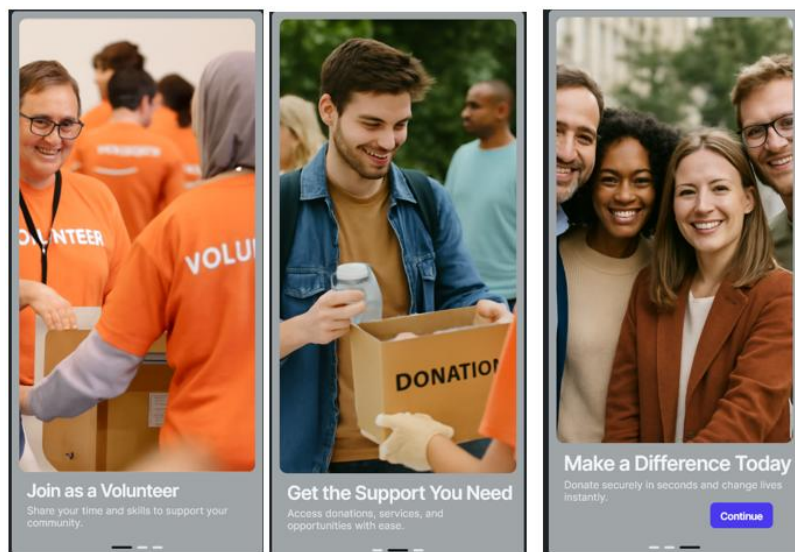


Figure 5.2- 2. OnBoarding

Figure 5.2-2 indicates that the onboarding screen helps users understand the main purpose and features of the AWN platform. The app shows you some of its key functions. You can volunteer, support community initiatives and donate to active campaigns. This gives you a sense of the use of the app before heading to the main app page.

5.3 User Role:

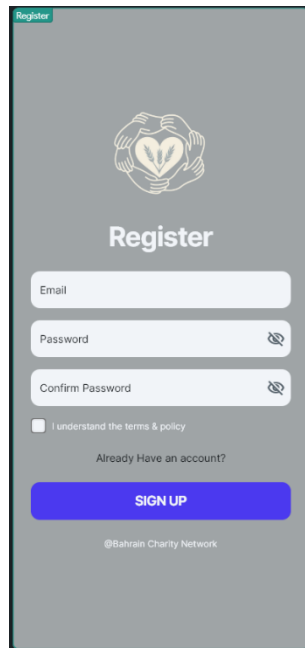
The image shows a mobile application registration screen. At the top left, there is a small green tab labeled 'Register'. In the center, there is a logo consisting of two hands holding a heart. Below the logo, the word 'Register' is displayed in a bold, sans-serif font. The form contains three input fields: 'Email', 'Password', and 'Confirm Password'. The 'Password' and 'Confirm Password' fields have eye icons to toggle visibility. Below these fields is a checkbox with the text 'I understand the terms & policy'. Underneath the checkbox is the text 'Already Have an account?'. At the bottom of the form is a large blue button with the text 'SIGN UP'. At the very bottom of the screen, there is a small text string '@Bahrain Charity Network'.

Figure 5.3- 1. Registration

Figure 5.3-1 shows that the registration screen allows a new user to register an account before using the system. To become a registered user on the website, a person will have to give their email ID and create a password. Further to register, they need to confirm the password and accept the terms and conditions. When users register, they become part of the Bahrain charity ecosystem and can start using donation, volunteering community and other AWN features.

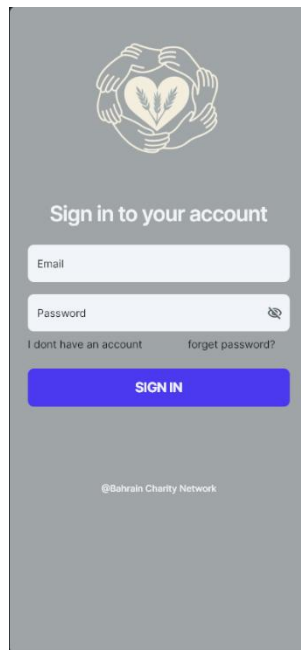


Figure 5.3- 2. Sign in

As we can see in Figure 5.3-3, the Sign In screen lets existing users access the AWN platform using their email and password. The screen also lets you recover your password if you forget it. Then there is another clickable item that redirects users to the registration screen if you do not have an account.

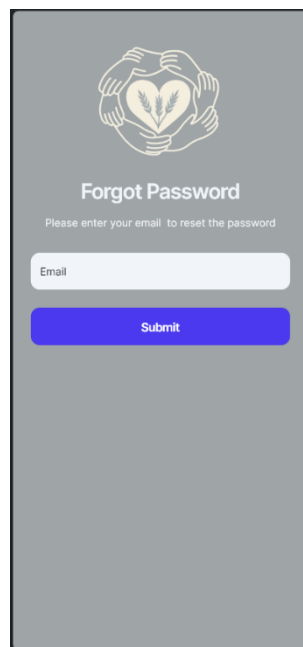


Figure 5.3- 3. Forgot Password

The Forgot Password screen allows you to reset your account password as shown in the screen of Figure 5.3- 3. When users submit their registered email address, they receive a recovery link that helps them to restore access and continue on the AWN platform.

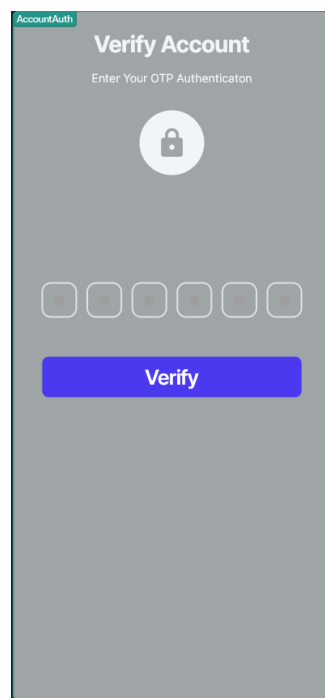


Figure 5.3- 4. Verification

According to figure 5.3- 4, this screen enables a secure activation of the account, as it requires the user to input the one-time verification code (OTP) sent to their email. When the right code is submitted, the account is verified successfully. The user can then continue to use the AWN platform hassle-free.

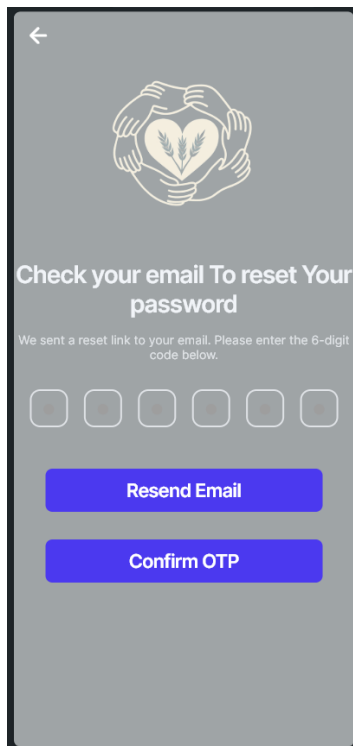


Figure 5.3- 5. Password Reset Verification

As shown in Figure 5.3-5, the system presents the Password Reset Verification screen when a user wants to recover their account. A 6-digit verification code will be sent to your registered email id which you must enter in order to set a new password. If you did not receive the code, then you are provided with an option on the screen to resend the code.

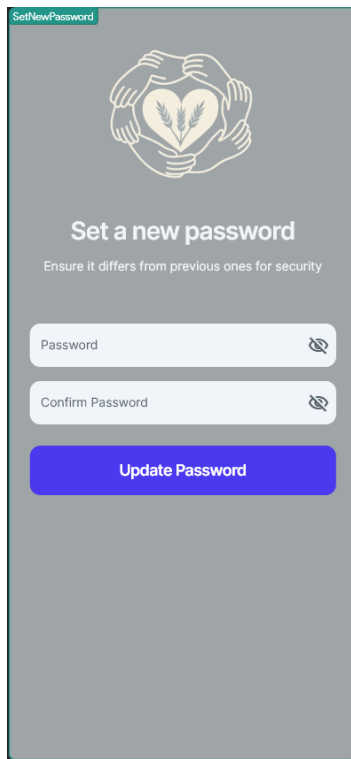
The image shows a mobile application screen titled "SetNewPassword" in a small green box at the top left. The background is a solid grey. At the top center is a circular logo consisting of several hands holding a heart, with a wheat stalk in the center. Below the logo, the text "Set a new password" is displayed in a bold, white font. Underneath this, in a smaller white font, is the instruction "Ensure it differs from previous ones for security". There are two white input fields with rounded corners. The first is labeled "Password" and the second is labeled "Confirm Password". Each input field has a small eye icon on the right side to toggle visibility. Below the input fields is a prominent blue button with the text "Update Password" in white.

Figure 5.3- 6. Set New Password

The screen shown in Figure 5.3-6 displays after the OTP process has been completed successfully, allowing the user to set a new password. The new password must be entered twice by the users. We will restore access to your AWN account as soon as possible after verification.

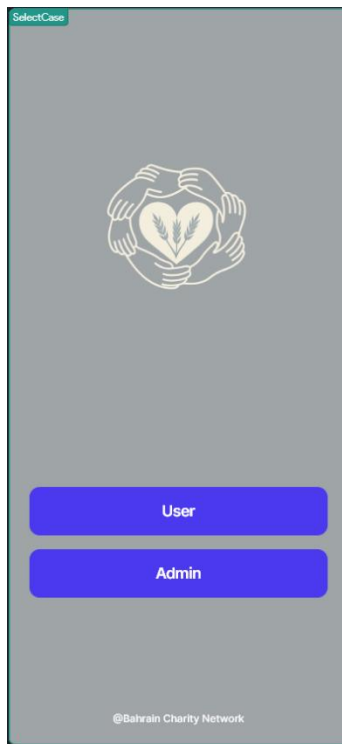


Figure 5.3- 7. Role Selection

As shown in Figure 5.3-7, the Role Selection screen appears before the sign-in process and allows individuals to select whether they will use the platform as a regular User or an Admin. This ensures that each user type is navigated to the correct login flow and receives the appropriate dashboard, permissions, and functionality within the AWN platform.

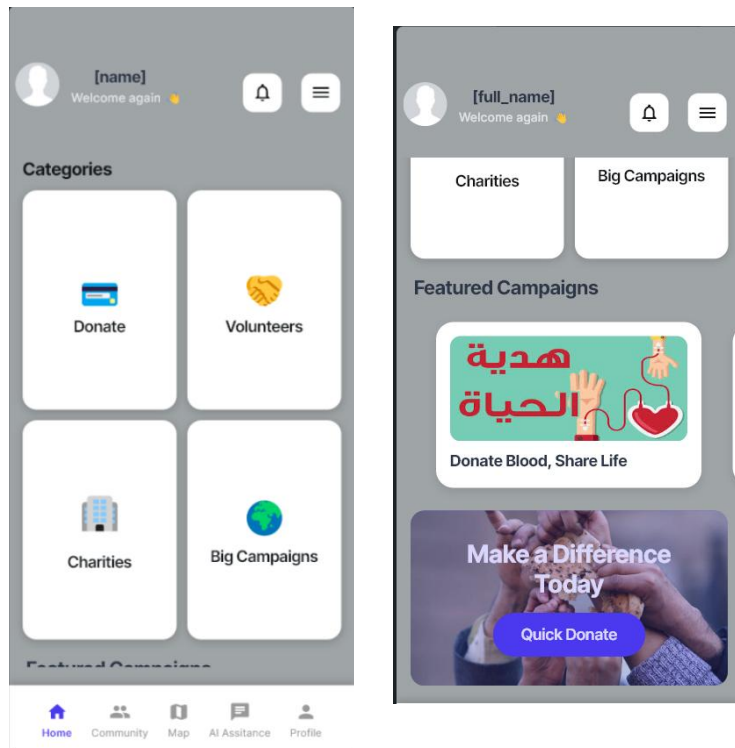


Figure 5.3- 8. Home page

Figure 5.3-8 shows us how the Home Page is the main screen users, after logging into the application. A welcome message along with quick access icons for notifications and the side menu is displayed on the interface. The page has some navigation cards such as Donate, Volunteers, Charities and Big Campaigns which take you to major sections. Highlighted campaigns also encourage donations for ongoing donation campaigns. At the bottom of the platform, there is a navigation bar that directs you to essential sections like Home, Community, Map, AI Assistant, and Profile.

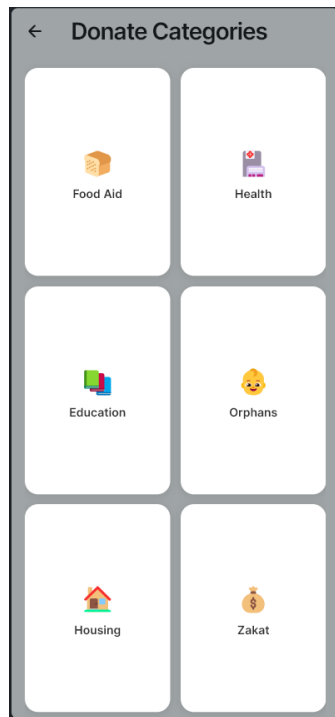


Figure 5.3- 9. Donate Categories

As shown in Figure 5.3-9, after selecting the option on the home page, the donate categories screen appears. The page allows users to mix and match various donation option categories such as Food Aid, Health, Education, Orphans, and Housing and Zakat. The site breaks down donations into sections, so donors will know where their money is going. Users can donate to specific causes.

← [icon][d_type]

[description]

Donation Amount

BHD 0.00

Payment Method

ApplePay  ☒

BenefitPay  ☐

Credit Card  ☐

Donate Now

Figure 5.3- 10. Donation Details

As can be seen in figure 5.3-10, when the user selects donation category, the Donation Details screen appears. It shows the type of donation chosen by the user and allows the user to enter an amount and description. You can use many payment options including Apple Pay, BenefitPay, and Credit Card for your convenience. When a user reviews the details, they can go ahead to click the Make a Donation Now button to contribute.

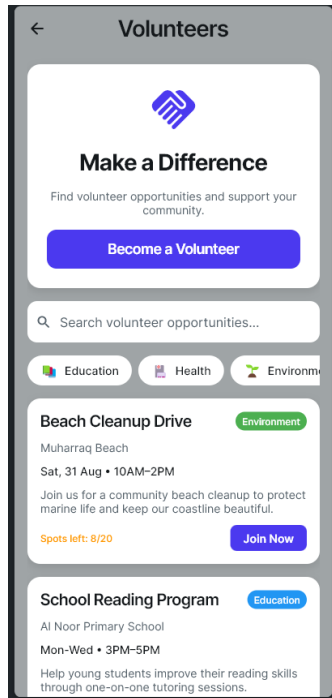


Figure 5.3- 11. Volunteers

As shown in Figure 5.3-11, the Volunteers screen allows users to explore available volunteering fields and select areas they are interested in supporting. This feature connects charities with suitable volunteers by matching interests, availability, and skills, making it easier for organizations to find the right support and for users to participate in meaningful community service.

← Become Volunteering

Your Information

Full Name

Email

Phone

Emergency Contact (Optional)

Notes

Preferences & Consent

Time Slot

Morning ☒ Night ☐

Skills

Driving First Aid

Teaching Translating

Account

I agree to event guidelines ☐

Become Volunteer

Figure 5.3- 12. Become Volunteering

As shown in Figure 5.3- 12, the Become Volunteering page allows users to register as a Volunteer by filling in the required information. The form starts with personal details which include a full name, email, phone number and emergency contact optional field along with a notes section for comments. The next part of the screen focuses on user choices like availability where the user can choose their preferred volunteering time slot (Morning or Night) and applicable skills. The user can tap on tags like Driving, First Aid, Teacher, Translator, etc. to choose applicable skills. Before submitting, there is a consent checkbox included to agree with the event guidelines. When users finish it, they can complete their registration by clicking Become Volunteer.

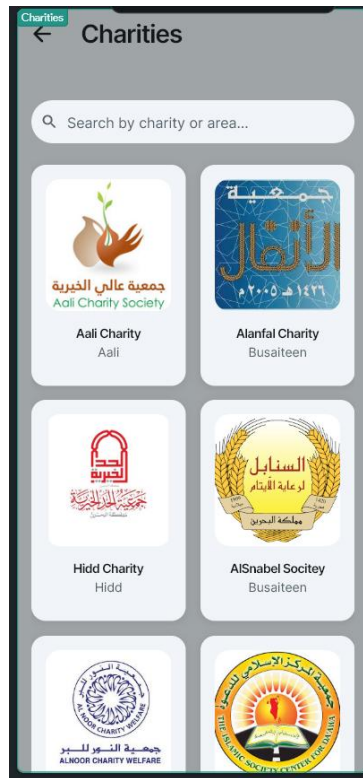


Figure 5.3- 13. Charities screen

According to image 5.3-13, the charities screen allows search and browse verified charities in Bahrain. Every charity card has the name, logo and location of the charity. This lets the user select a charity easily and donate. A bar at the top lets users filter and find organizations quickly. This screen helps the user see and search charities they can trust before making the choice to donate or volunteer.

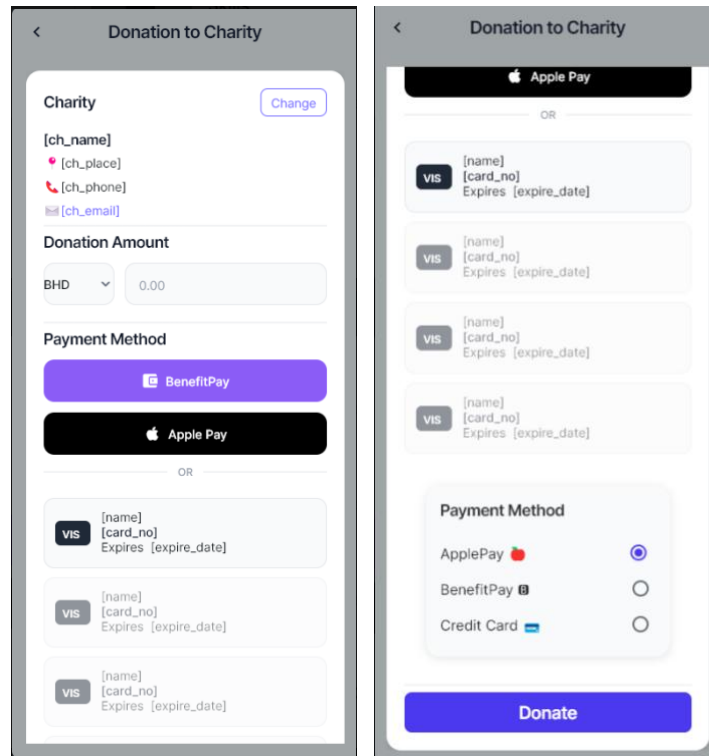


Figure 5.3- 14. Inside Charities

As seen in Figure 5.3- 14, users are able to donate to a selected charity of their choice, along with the charity's name, location, phone number, and email for verification purposes on the Inside Charities page. The user may enter the donation amount in BHD and choose from a variety of secure payment methods including BenefitPay, Apple Pay, or saved credit cards. When the user selects the preferred method, the process is completed by the user pressing the Donate button. Thus, the transaction is easy, quick, and user-friendly.

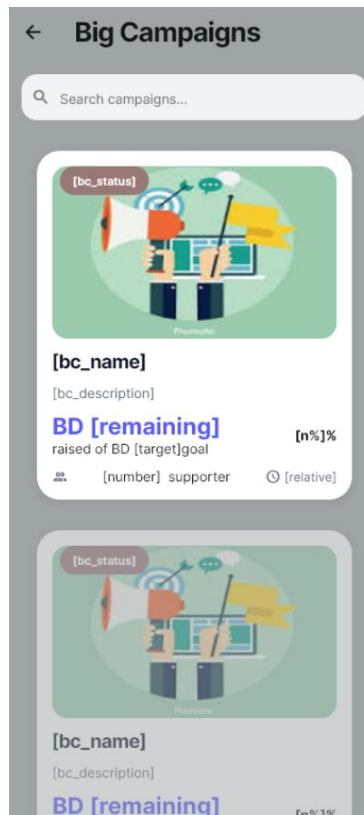


Figure 5.3- 15. Big Campaigns

Figure 5.3- 15 demonstrates the Big Campaigns page. Large-scale charity campaigns take place in this section. Campaigns display card views that contain the following details: name, description, target remaining balance and total amount of campaign donations. Each campaign card tells you how many percent of the campaign is complete and how many have already supported the project to get a sense of urgency or impact. There is a search bar on top of the page to help users find specific campaigns easily. As such, it makes browsing, tracking and selecting donation opportunities easy.

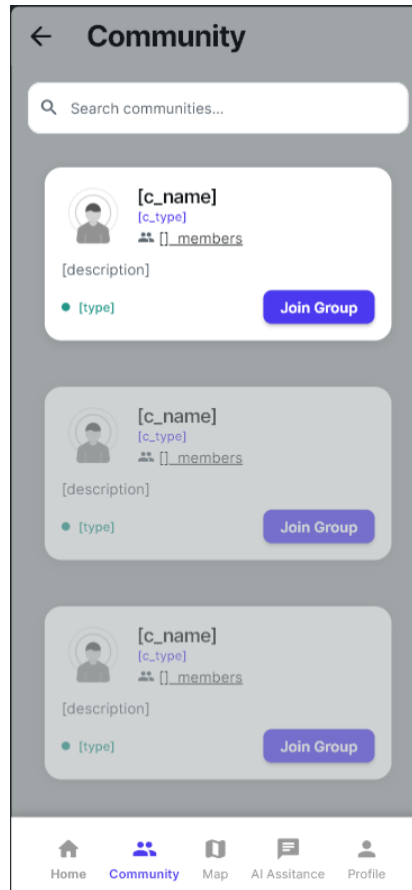


Figure 5.3- 16. Community

As shown in Figure 5.3- 16, a Community Page is available on the AWN platform. Through this page, users can join groups formed either on the basis of particular interests or based on a charity of their choice. The community’s name, category type, description and number of current members are displayed on each group card. At the top, a search bar allows users to quickly filter and find communities based on their interests. The “Join Group” button makes it easy for users to join a group, encouraging people to work together and communicate while also boosting involvement in charity.

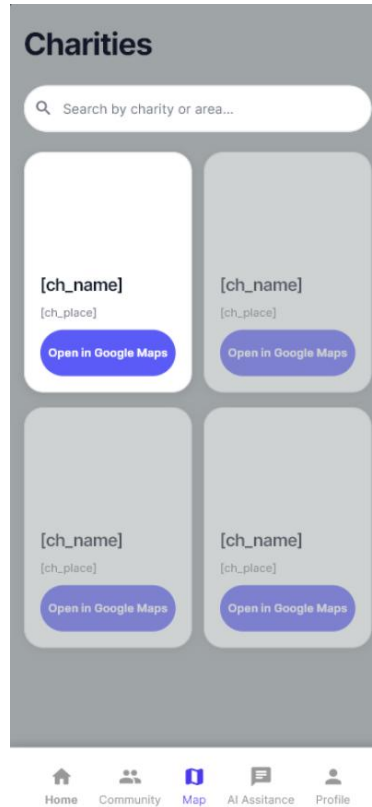


Figure 5.3- 17. Map Page

According to Figure 5.3- 17, the Map Page allows the user to find nearby charities based on name or geographic area. The search filter on top makes it easy for users to find relevant organizations. Every charity card shows the charity name and location. There is also an “Open in Google Maps” button which enables the user to instantly get directions to the charity or visit the charity in person. It improves access and connects donors, volunteers and charities throughout Bahrain.

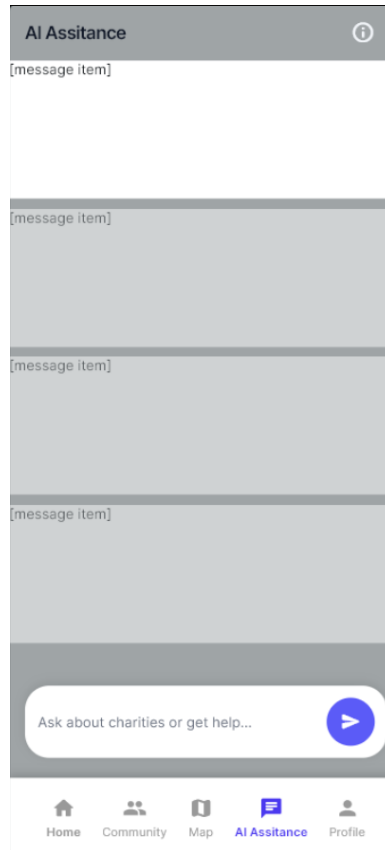


Figure 5.3- 18. Ai Assistance

According to Figure 5.3- 18, the AI Assistance page is a place where one can ask questions regarding charities and donation and volunteer opportunities or general help regarding the platform. The screen shows messages in a chat-style, as previously seen, so you can think of the interaction as being natural. A text input field at the bottom lets users type a query and send it through the button without any delay. This feature helps the user to understand what they should do next preventing confusion and increasing overall usability.

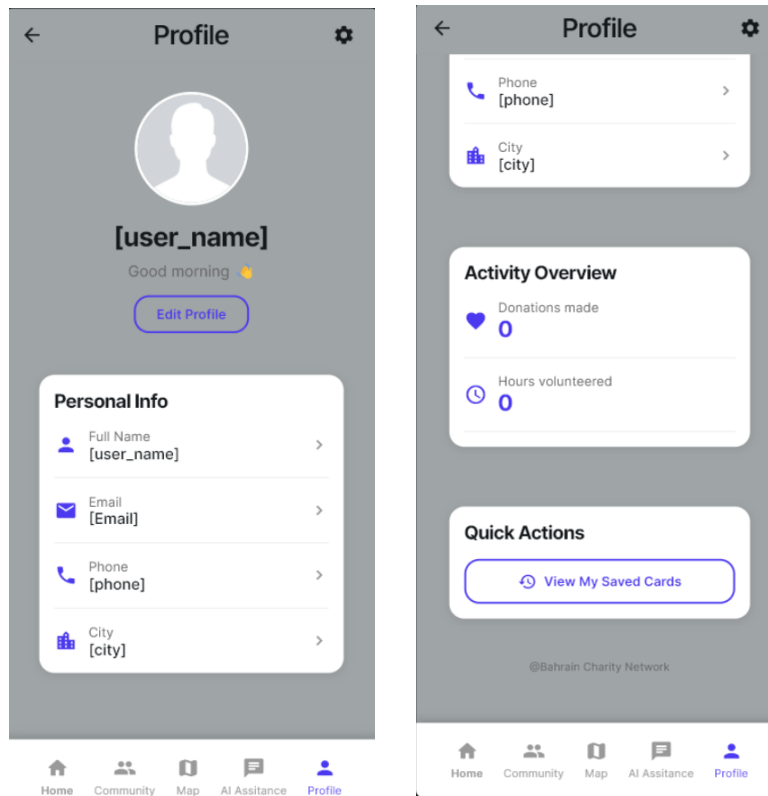


Figure 5.3- 19. Profile Page

According to Figure 5.3- 19, the Profile Page is where users can view and edit their information on the system. The page shows your name, email, phone, and city. It also features an edit profile option. Moreover, there is an Activity Overview section below. This section will show the total donations and hours offered by the user. The Quick Actions section also simplifies things by providing saved payment cards for faster transactions in the future. This page organizes everything about your account and participation history. You can also update your details from here and keep your data secure.

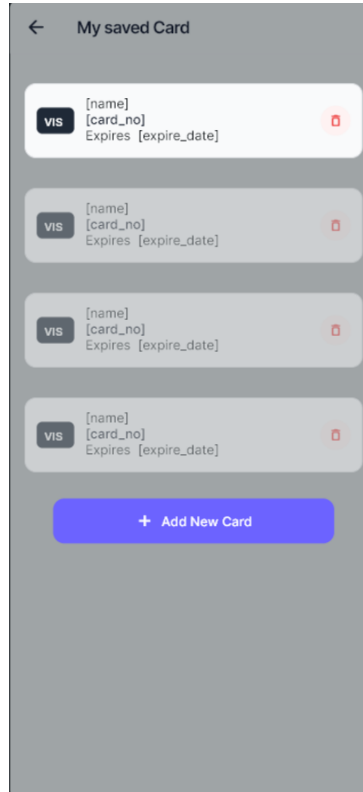


Figure 5.3- 20. My saved card

According to Figure 5.3- 20, on the My Saved Card page, users can manage their saved cards in a secured environment. The name, part number, and expiry will be shown for every saved card so that you are clear without any revealing of card details. Besides every card, there is a delete option that allows the users to either delete their unused payment methods or anything that is not in use. In addition, this page contains an "Add New Card" button to register another payment option to make future donations quickly and conveniently.

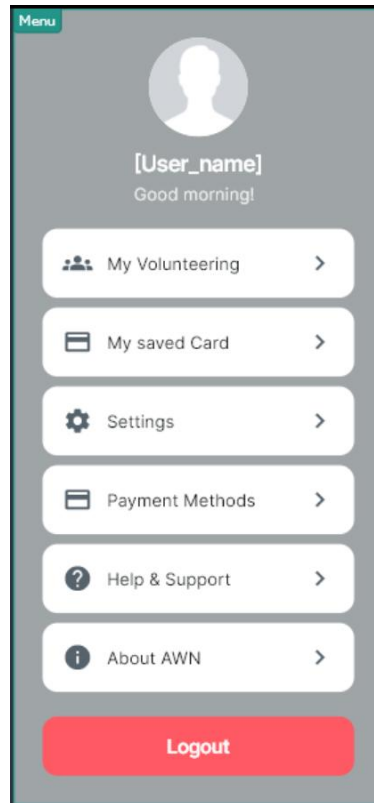


Figure 5.3- 21. Menu Component

According to Figure 5.3- 21, the different menu components of the application provide easy access to key features and personal settings of the user. This menu has My Volunteering, My Saved Card, Settings, Payment Method, Help and Support, About AWN etc so that users can easily access whichever option they want. Further, the interface has a logout button visible at the bottom which could help the user log out quickly and safely. The layout will be simple, orderly and straightforward. It enables users to access relevant tools and information with the least possible effort.

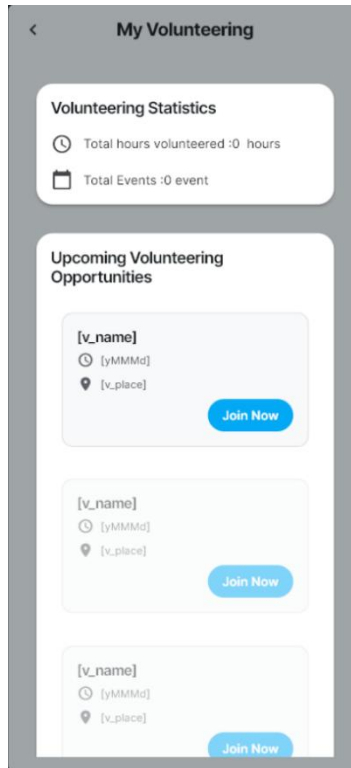


Figure 5.3- 22. My Volunteering

Figure 5.3- 22 shows my volunteering page which helps to keep a record of volunteering and to be notified of upcoming events. The upper section consists of volunteering statistics providing total hours and completed events, making it easy for users to navigate their engagement. Users can check the available volunteering activities below along with other details like event name, date and location. Each opportunity that appears on the website has a “Join Now” button that lets you quickly sign into the community. This page motivates, informs and updates users about upcoming volunteer opportunities in the future.

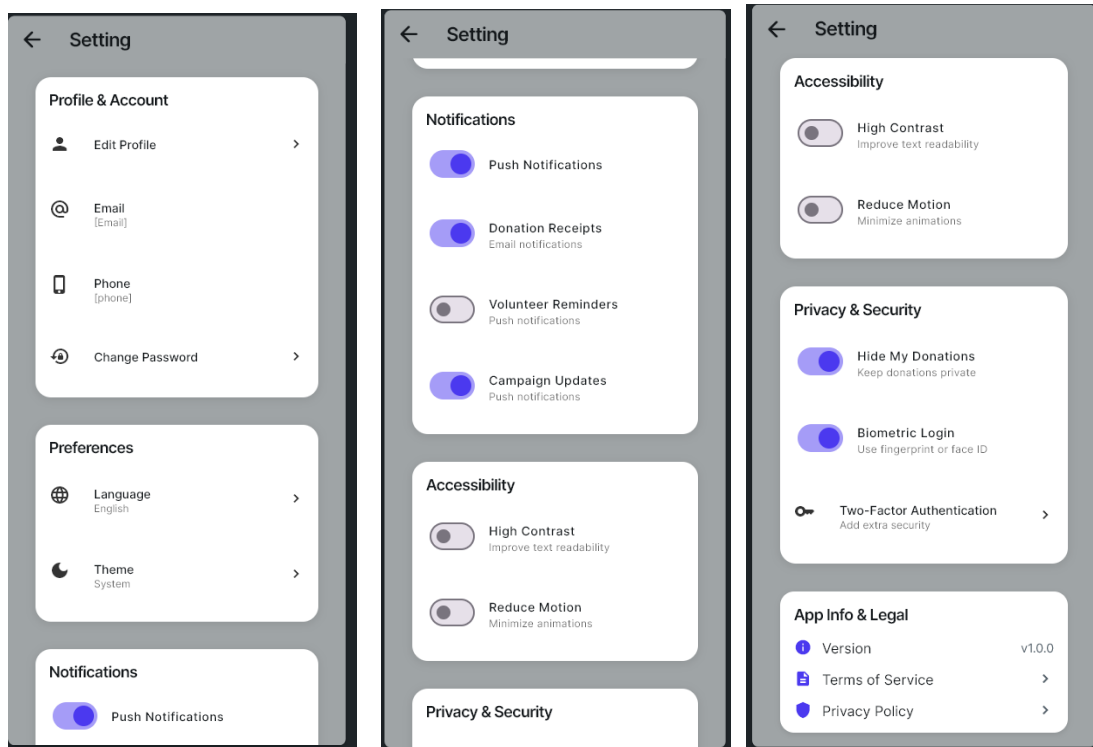


Figure 5.3-23. Settings Page

Figure 5.3- 23 shows the Settings page that allows users to personalize every aspect from account information to privacy option within the AWN app. On the page, there are sections provided for editing personal information such as email, phone number, and password. Language and theme preferences for better accessibility are also there. The users can also manage notification types like donation receipts, campaign reminders, and volunteer alerts. You can also enable high contrast mode, turn off animations, use biometric login and two-step signing for better use. This settings page allows each user to set the app experience to their own comfort, access needs and privacy preferences.

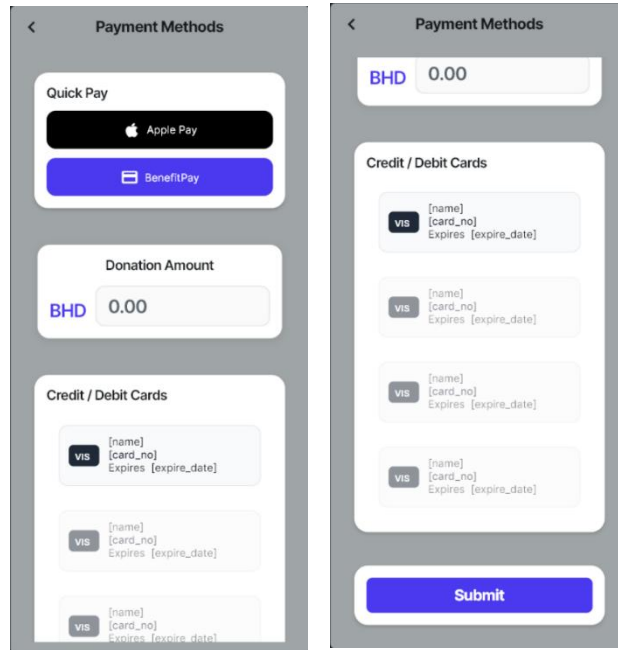


Figure 5.3- 24. Payment Methods

Figure 5.3- 24 shows the Payment Methods page where users can safely manage their donation payment options within the AWN application. The interface gives a convenient approach for quick payments using fast-payment Apple Pay and BenefitPay payment solutions. Users may also enter an amount manually and choose any of their saved credit or debit cards for their recurring or future payments. The arrangement of the layout together with the confirmation button will assist you in avoiding error in the donation. Furthermore, the donation process is seamless, flexible, and user-friendly.

The screenshot shows a mobile application interface for adding a new payment card. The title bar at the top is grey with a back arrow on the left and the text 'Add New Card' in the center. Below the title bar is a white card with the heading 'Card Information'. The card contains four input fields: 'Cardholder Name' with a placeholder 'Enter cardholder name', 'Card Number' with the value '1234 5678 9012 3456' and a character count '0/16', 'Expiry Date' with a placeholder 'MM/YY' and a character count '0/5', and 'CVV' with the value '123', a character count '0/3', and a small eye icon to toggle visibility. At the bottom of the white card is a blue button with the text 'Save Card'.

Figure 5.3- 25. Payment Methods

As can be seen in Figure 5.3- 25, the Add New Card page enables the user to safely save a new payment card in the app for future donations. The form pulls the cardholder’s name, credit card number, expiry date, and cvv for validation at the top of the page. After the card is completed, the user clicks the “Save Card” button to store the card which allows a quicker and easier checkout, saving payment information for better transactions.

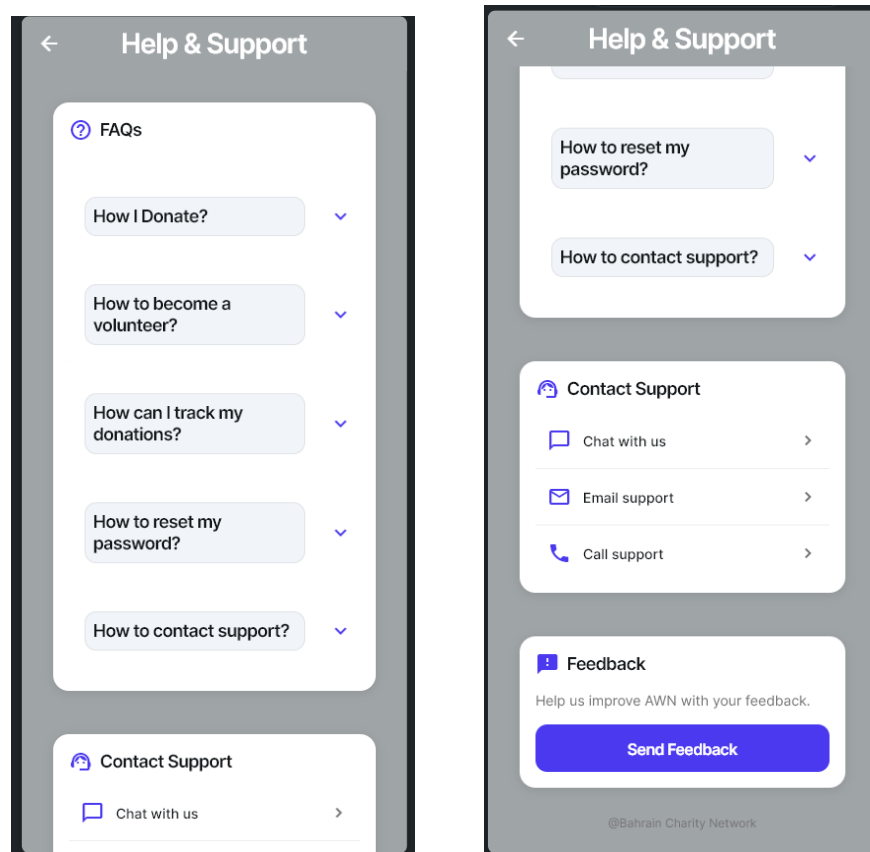


Figure 5.3- 26. Help & Support

As depicted in Figure 5.3- 26, the Help & Support page offers users a quick way to get answers and help when needed. The page has a list of expandable FAQs for direct guidance. It also has live chat, email and call support for further assistance. Moreover, users can submit their suggestions on the feedback section, which helps to enhance the improved version of AWN application.



Figure 5.3- 27. About AWN

Figure 5.3-27 shows the AWN about page which gives the respectable purpose of the site. AWN aspires to consolidate donation, volunteering, and aid services in an easy-to-use platform that supports all verified charities across Bahrain. The platform puts forward its values like trust, transparency and safety, assuring users that only vetted charities are available on it. The page may also contain impact stats, contact information, and legal information of the AWN which creates connectivity between donors, volunteers, and organizations for social impact.

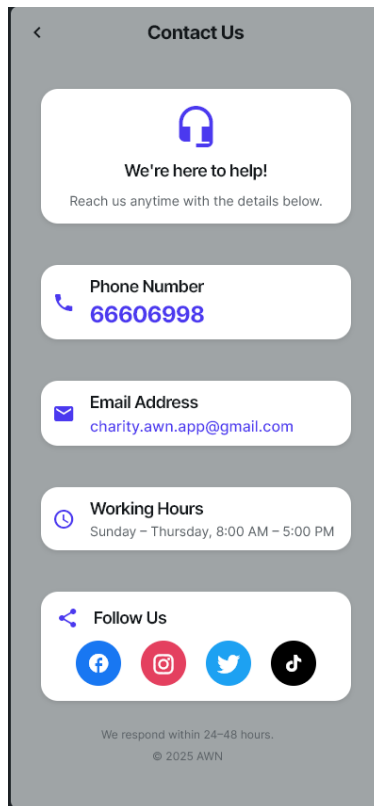


Figure 5.3- 28. Contact Us

Contact Us screen shown in Figure 5.3-28 provides a way for customers to communicate with the AWN support anytime help is needed. It shows things like the official email address, a hotline number, and its operating availability. Also, links to social sites like Facebook, Instagram, Twitter, and TikTok provide updates on announcements, events, and charity initiatives throughout Bahrain. This ensures that users can easily connect with users.

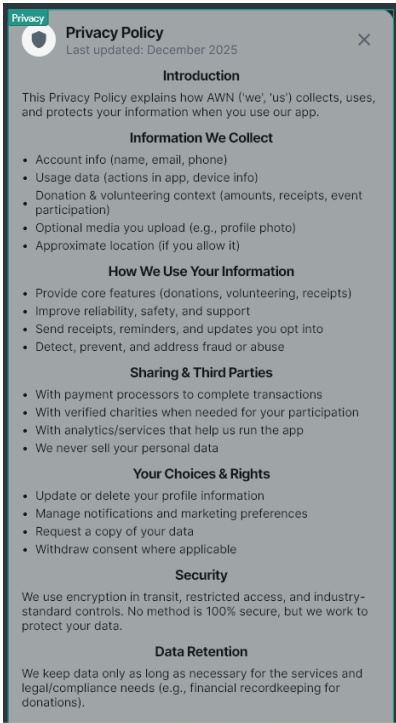


Figure 5.3- 29. Privacy Policy

As shown in Figure 5.3-29, the Privacy Policy screen in the AWN Charity App shows how the app collects and stores user data. The policy outlines what information is collected such as details of your account, past donations, optional geolocation access, user setting. It also explains how this data is used, specifically to improve app functionality, facilitate secure transactions, support verified charities, and enhance user experience. Also, users are informed about their rights which include being able to access, update or request the removal of their data. This page reinforces AWN’s commitment to protecting the privacy of users and data and sensibly handling sensitive information in Bahrain charity.

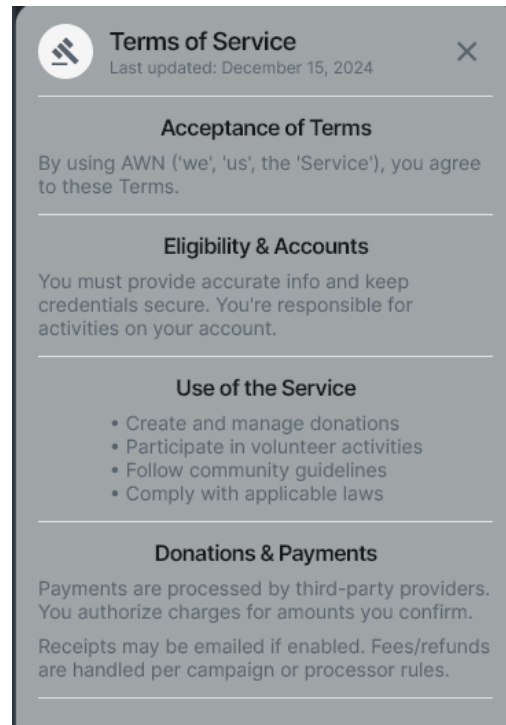


Figure 5.3- 30. Terms of Service Component

Figure 5.3-30 shows the Terms of Service screen displaying the official usage guidelines for the Awn Charity App. When creating an account or donating, applying for volunteer positions, or using community features, users must comply with the user liability guidelines. The terms also highlight following the laws and ethical standards of Bahrain, behaving respectfully, and using the services properly. Also, it makes sure to let you know that all transactions processed by trusted third-party payment providers are safe and transparent and fully accountable. This ensures users understand their rights and obligations for the use of Awn and that AFN operates in a safe, reliable, and regulated environment for charities.

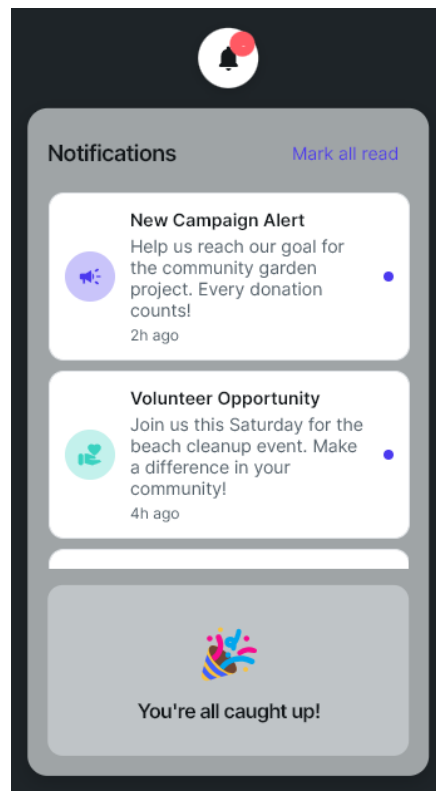


Figure 5.3- 31. Notifications Component

Figure 5.3- 31 shows the Notifications component that appears with important news like New Campaign, Volunteering Opportunities and System Alert. Users can track the unread messages easily and also use the option to mark all as read. When the user sees the notifications, it displays the user is completely up to date.

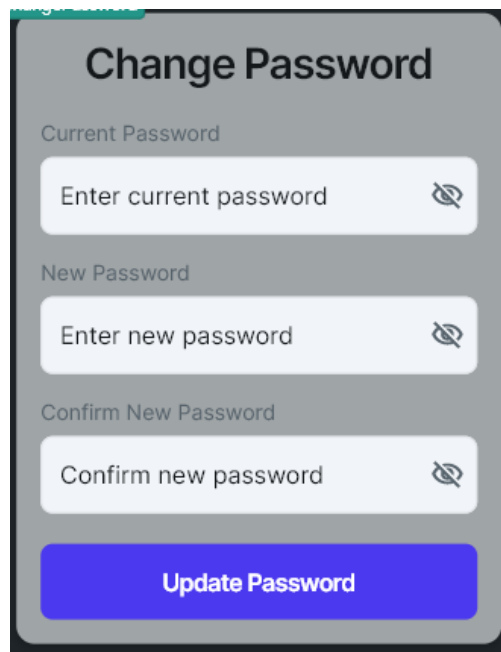
The image shows a mobile app screen titled "Change Password". It features three input fields for password entry, each with a placeholder text and a toggle icon. The first field is labeled "Current Password" with the placeholder "Enter current password". The second field is labeled "New Password" with the placeholder "Enter new password". The third field is labeled "Confirm New Password" with the placeholder "Confirm new password". Below these fields is a large blue button labeled "Update Password". The entire form is set against a light gray background with rounded corners.

Figure 5.3- 32. Change Password Component

According to figure 5.3-32, the Change Password screen within the AWN Charity App allows users to change their login credentials securely. To do this, you need to enter your present password, then new password and confirm it. This feature maintains security, it gives the user complete control over their authentication details. It makes logging into your profile, donations and volunteering much safer.

← Send Feedback

How would you rate your experience?

★★★★★

1 - Very poor, 2 - Poor, 3 - Average, 4 - Good, 5 - Excellent

Category

Bug Feature Payment

Volunteering Account

Content Other

Details

Describe your issue or idea in detail...

Minimum 10 characters required

Send Feedback

Figure 5.3- 33. Send Feedback

As depicted in Figure 5.3- 33, the Send Feedback page enables users to report their experience along with any problems or suggestions. This form has a star rating and a few categories to select from along with an enter your detail text field. When you're finished, you can give your feedback to help us improve the platform and future users.

5.4 Admin Role:

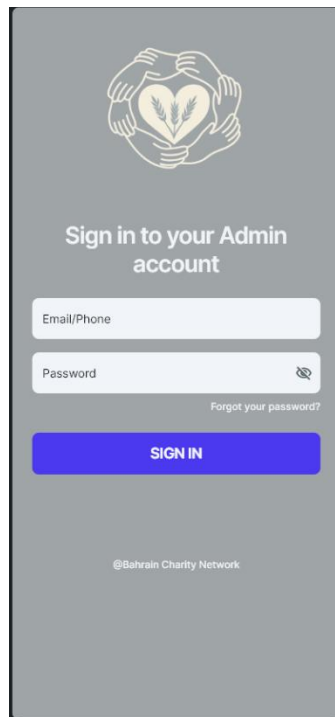


Figure 5.4- 1. Admin Login Page

As shown in Figure 5.4-1, this screen represents the Admin Login Page of the AWN Charity App. Administrators can securely access their accounts by entering their registered email or phone number along with a password. A "Forgot Password" option is also available to assist administrators in recovering access when needed. This page serves as the entry point for authorized personnel to manage platform operations, oversee charity records, and maintain system integrity within the Bahrain Charity Network.

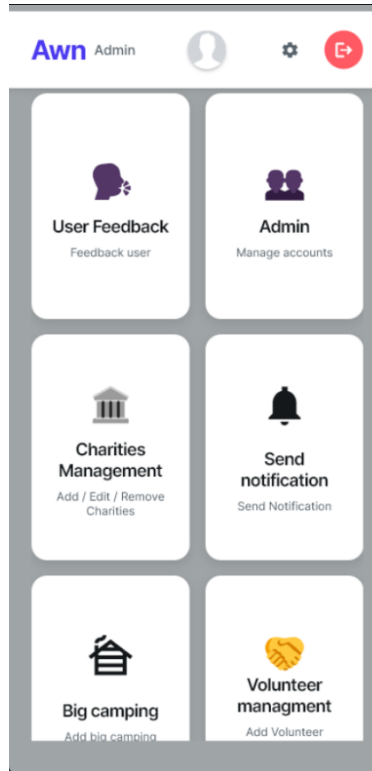


Figure 5.4- 2. Admin Home Page

According to Figure 5.4- 2, the Admin Home Page is a dashboard that allows the administrators to perform all major management functions on the Awn in one place. The interface has clearly labeled cards indicated with “user feedback”, “admin profile”, “edit/delete charity”, “send notify”, “big campaign”, “volunteer opportunities”. The upper segments comprise the admin’s profile and necessary navigation symbols, which ensures user settings and system controls are accessible. This design helps administrators to facilitate oversight of the platform operation and the management system efficiently.

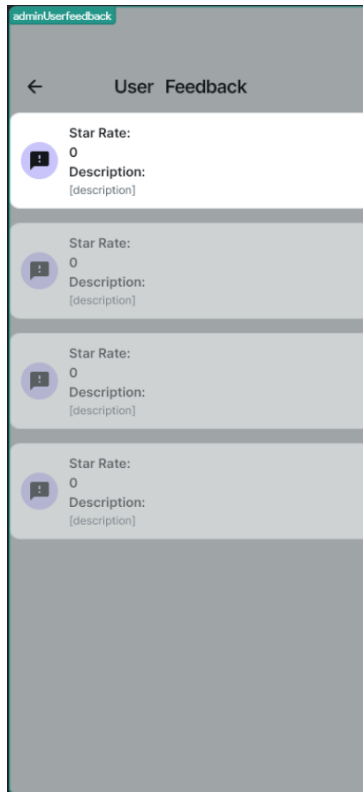


Figure 5.4- 3. User Feedback

As illustrated in Figure 5.4- 3. User Feedback, the User Feedback page enables the administrator to view all of the feedback provided by users. The administrator will receive each feedback card with the star rating accompanied by a description. Thus, it helps the administrator to know the user experience and areas of improvement. The layout is scrollable on the page which makes all entries clear and accessible. This screen helps to track how satisfied our users are. It allows us to gather data to continually improve the AWN platform.

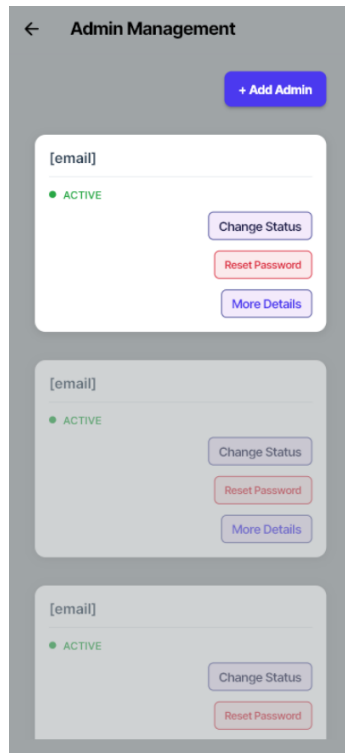
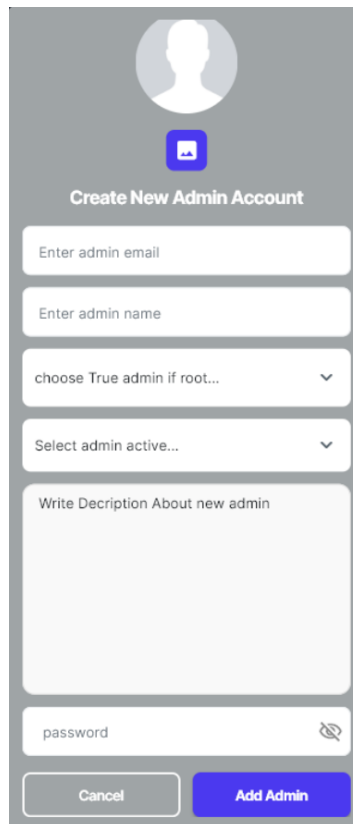


Figure 5.4- 4. Admin Management

As shown in Figure 5.4- 4, the Admin Management page allows authorized users to view and manage existing admin accounts. Each admin entry displays status information along with available actions such as changing access status, resetting passwords, or viewing additional details. A button at the top enables adding new administrators, making system governance flexible and controlled.



The image shows a mobile application interface for creating a new admin account. At the top, there is a grey header bar containing a white silhouette of a person's head and a blue square icon with a white document symbol. Below the header, the title "Create New Admin Account" is displayed in white text. The form consists of several input fields: "Enter admin email", "Enter admin name", a dropdown menu labeled "choose True admin if root...", another dropdown menu labeled "Select admin active...", a large text area labeled "Write Description About new admin", and a password field labeled "password" with a toggle icon. At the bottom, there are two buttons: a grey "Cancel" button and a blue "Add Admin" button.

Figure 5.4- 5. Add Admin Component

As you can see in Figure 5.4- 5, the Add Admin section of the system provides a form for the admin to create another admin. The interface consists of the required fields such as email, full name, role selection, account status - and a description field. You are given a password field to complete the setup. You also have action buttons like a cancel button and a confirm button for adding the new admin.

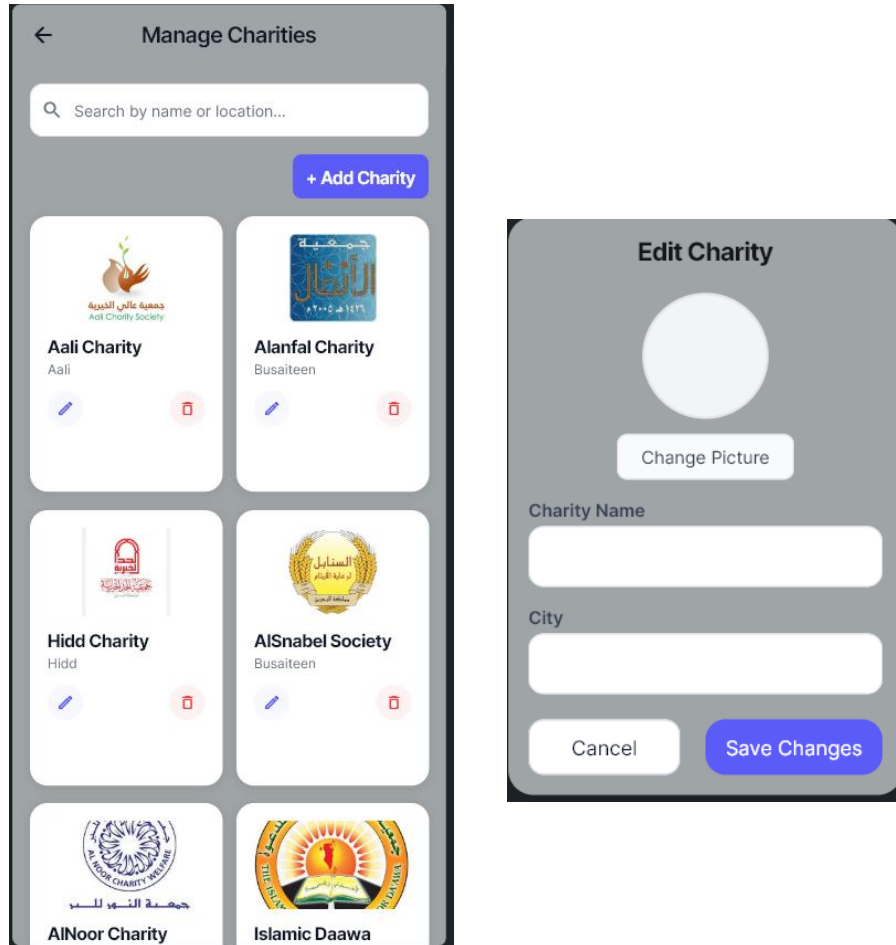
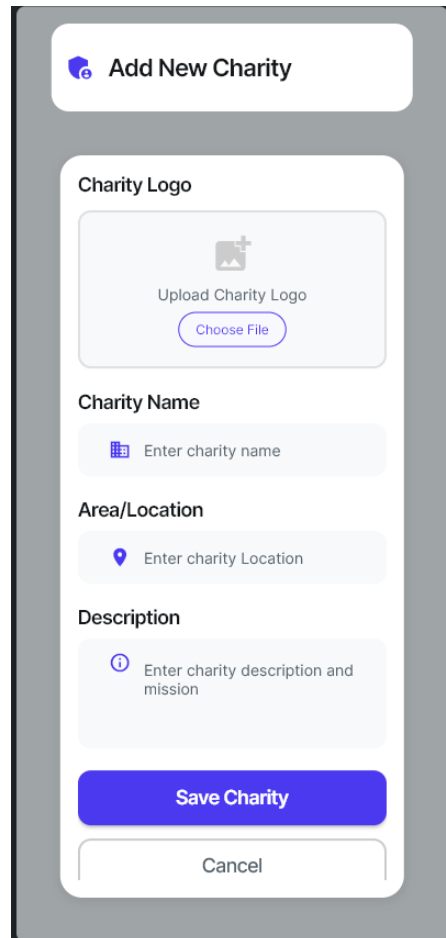


Figure 5.4- 6. Manage Charities Page

According to Figure 5.4- 6, the Manage Charities Page in Awn Admin represents this screen. Charity Administrators can view, add, edit, and remove the charity organizations from this interface. By selecting the edit icon, you can open the Edit Charity Component, which allows you to update the charity name, city, and logo. Clicking the delete icon opens a pop-up so you don't delete by accident. This short page outlines the most efficient and systematic way to maintain correct and updated charity information in the Bahrain Charity Network.



The image shows a mobile application interface for adding a new charity. At the top, there is a header bar with a purple icon and the text "Add New Charity". Below this, the form is organized into sections: "Charity Logo" with an "Upload Charity Logo" button and a "Choose File" link; "Charity Name" with a text input field; "Area/Location" with a text input field; and "Description" with a text input field. At the bottom, there are two buttons: a prominent purple "Save Charity" button and a white "Cancel" button.

Add New Charity

Charity Logo

Upload Charity Logo

Choose File

Charity Name

Enter charity name

Area/Location

Enter charity Location

Description

Enter charity description and mission

Save Charity

Cancel

Figure 5.4- 7. Add New Charity Page

According to Figure 5.4- 7, the Add New Charity page presents a form-based interface that helps administrators to register new charities in the system. The page will have a field to upload charity logo. Next, there will be a field to enter charity name. Further, location and a description to outline purpose and mission of the charity. The button to finalize the submission is called “save charity”. This has an option given which says “cancel”. Moreover, through which the administrator can go out without saving any changes.

← Send Notification

Notification Title

Enter notification title

Notification Type:

Select... ▼

Message Content

Enter your message here...

Send Notification

List of Notification

[n_type]
[notification_name]
[n_description]
[relative]

[n_type]
[notification_name]
[n_description]
[relative]

Figure 5.4- 8. Send Notification

According to figure 5.4- 8, the Send Notification page allows the administrator to create and send notifications to users. The interface consists of a title field to write the notification title, a dropdown menu field to choose the type, and an area to write the message. After finishing, the admin can send the notification by using primary action button. At the bottom of the form, a list of notifications sent previously will appear which an admin can track or remove or review the notifications that were sent previously.

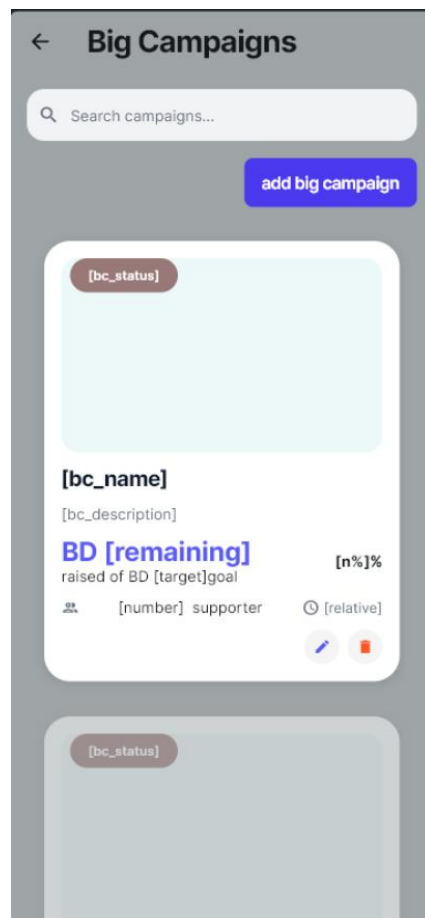


Figure 5.4- 9. Manage Big Campaigns

As shown in the figure 5.4- 9, admins can manage all active fundraising campaigns from AWN application via Manage Big Campaigns page. In the design, there is a search bar that quickly filters, plus a button to add campaigns. All the important information about the campaign card will be displayed including the title, description, the progress bar, and the count of charities or donors supporting. The icons are placed easily so that you can instantly edit or delete the campaign. The purpose of the AWN platform is to allow different charities to run large scale projects all in one place, making it easy to coordinate, track and manage.

The image displays two mobile application screens for creating a new campaign. The left screen, titled 'Add New Big Campaign', features a 'Campaign Logo' section with a light blue image placeholder and a 'Choose File' button. Below this are input fields for 'Campaign Name' (with a document icon), 'Campaign Target amount' (with a dollar sign icon), and a 'Campaign Description' text area with the placeholder 'Write Campaign Description'. The right screen shows a similar layout but with a larger image placeholder at the top, a 'Choose File' button, and a 'Save Big Campaign' button at the bottom, along with a 'Cancel' button.

Figure 5.4- 10. Add new Campaigns

According to Figure 5.4- 10, the Add New Campaign page formelt gives every administrator an opportunity to create and launch new fundraising campaigns. On this page, you will see space to upload campaign image, write the campaign name, define target amount for donation and write a detailed description of the campaign. Once the administrator provides all the required information, he can either save the new campaign or cancel it.

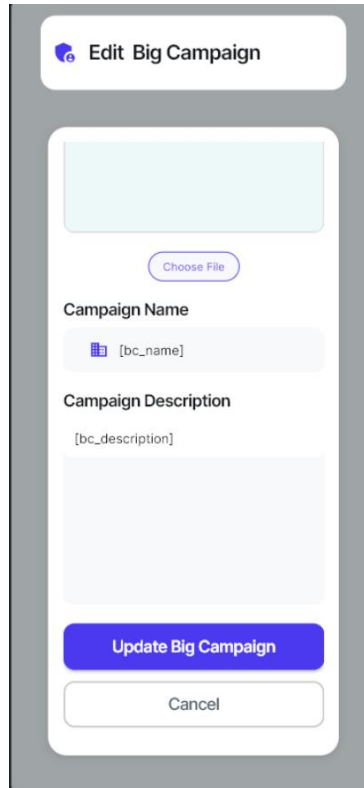
The image shows a mobile application interface for editing a campaign. At the top, there is a header bar with a purple icon and the text "Edit Big Campaign". Below this is a large light blue rectangular area, likely for a campaign image. Underneath the image area is a "Choose File" button. The form then has two text input fields: "Campaign Name" with a placeholder "[bc_name]" and "Campaign Description" with a placeholder "[bc_description]". At the bottom of the form are two buttons: a blue "Update Big Campaign" button and a white "Cancel" button with a grey border.

Figure 5.4- 11. Edit Campaign Component

As shown in Figure 5.4- 11, the Edit Campaign Component allows administrators to update existing campaign information easily. The page provides editable fields to modify the campaign name, update the campaign description, and change the campaign image if needed. After making the necessary adjustments, the administrator can apply changes by selecting the “Update Big Campaign” button or cancel the action to return without updating.

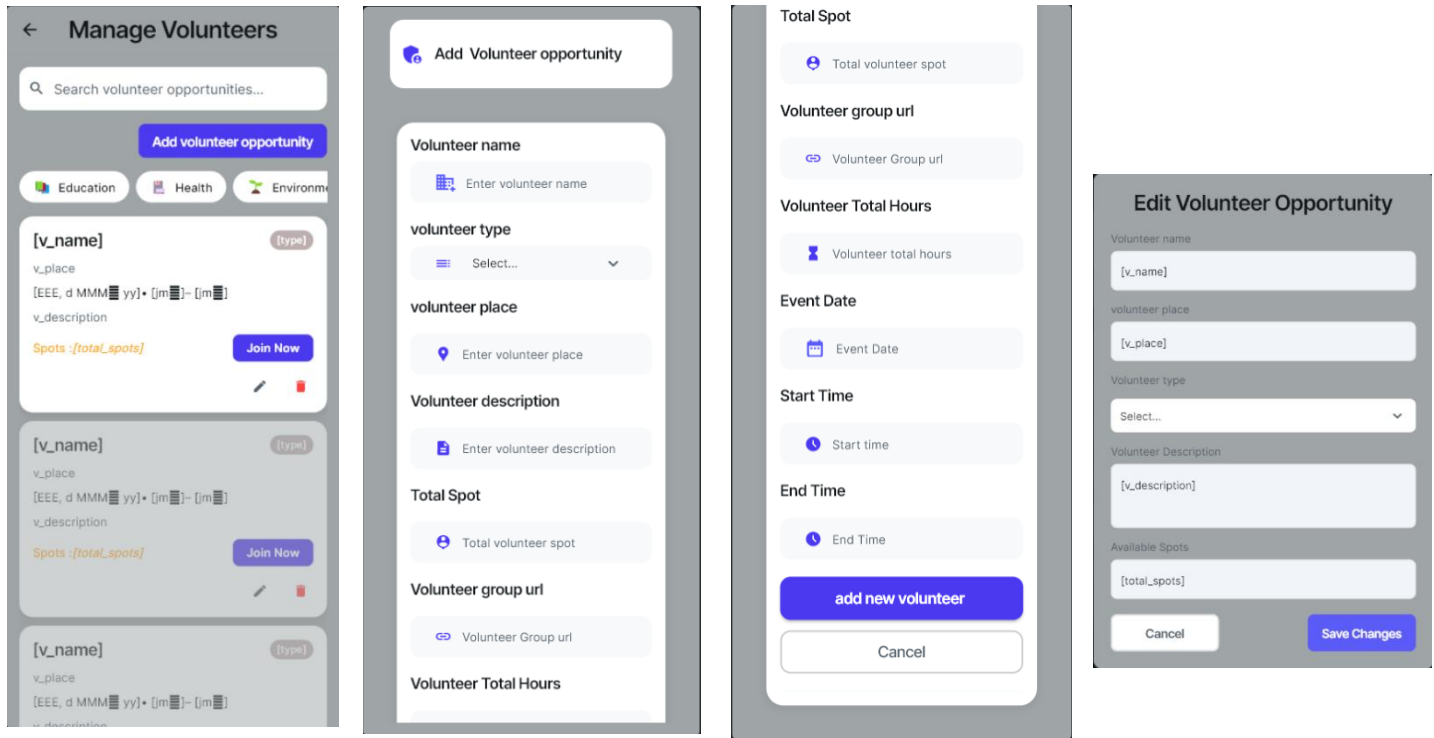


Figure 5.4- 12. Manage Volunteers

As depicted in Figure 5.4- 12, Manage Volunteers, users are able to view all available volunteer opportunities from one central location. Users can see a list of opportunities that contain name, location, description, date, and number of remaining spots. Using a specific form which includes all the fields, they can create more opportunities to feed the required data. Also providing the admin an editing page to update any existing opportunity easily. The admin has the option to save after changes or cancel and go back without changing anything.

Chapter 6

System Implementation and Testing

6.1 Introduction

The purpose of implementation and testing is to transform the design into a working system of Awn Charity App which will connect Bahrain's donors, volunteers, charities and administrators. We built the core parts like authentication, donations, volunteer affairs, and notification on FlutterFlow for front end and Supabase for backend service. When the implementation is over, various testing is conducted which includes Functional Testing, Usability Testing, Performance Testing, Bug fixing, and much more to ensure that the application offers the desired output. Through testing activities, the reliability of the system and validation of the application were confirmed that the application would operate smoothly and provide a user-friendly experience as required.

6.2 Tools Used in This Project

The latest cloud-based technologies have been exploited to build the Awn Charity App. We made sure the app had effective implementation & secure data handling with seamless integration of the user interface and backend. We used FlutterFlow for frontend and Supabase for backend services such as authentication and database services.

6.2.1 FlutterFlow

The FlutterFlow app was used for the building front-end and overall User Experience of Awn Charity App. With the drag and drop components, widgets that responded to conditions and logic flows, a responsive interface could be created. Using FlutterFlow, we configured navigation control, form validation, role-based access, and integration with external services to quickly develop registration screens, donation processes, volunteer screens, account settings, and community functionality.

6.2.2 Supabase

For the AWN application, I used Supabase to manage authentication, store data and carry the communication between the app and the database. AWN implemented email verification, password reset, OTP verification, and role-based permissions for users, charities, and admin thanks to Supabase. Through Supabase, we stored and managed all structured data such as user profiles, donations, records of volunteering, charity information, and big campaign information. Because of its real-time synchronization, any updates made would immediately appear on the AWN interface in FlutterFlow. Similarly, the donation progress, campaign, and volunteer activity were always current.

6.3 Implementation

The implementation phase includes transforming the comprehensive structural and design plans of a system into a full-scale application. The development was done step by step to make sure each feature works properly, is connected to the backend, and conforms to the user requirement. During this phase we implemented functional module like Authentication, Database connectivity, Donation, Volunteer, navigable flows & profile customizations etc. The connection between FlutterFlow and Supabase allowed live data handling and secure user access. Moreover, every feature worked consistently across different types of users. They altered what was in the system multiple times as the design got changed, logic flow problems were fixed and the system did what it was meant to do.

6.3.1 Database Structure

The core data of the application is stored using the Supabase database structure. We created tables for users, charities, donations, campaigns, volunteers, saved campaigns, payments, etc. and created the required database. To ensure uniformity, correctness as well as integrity of data, each table was defined with suitable field types and identifications. To help the features of the system, the relationships have been logically created between the tables. For example, a user has been linked to donation, volunteer has been linked to the volunteering events and user is linked to saved

campaign, etc. This structure approach made it sure that the data operations were efficient and scalable.

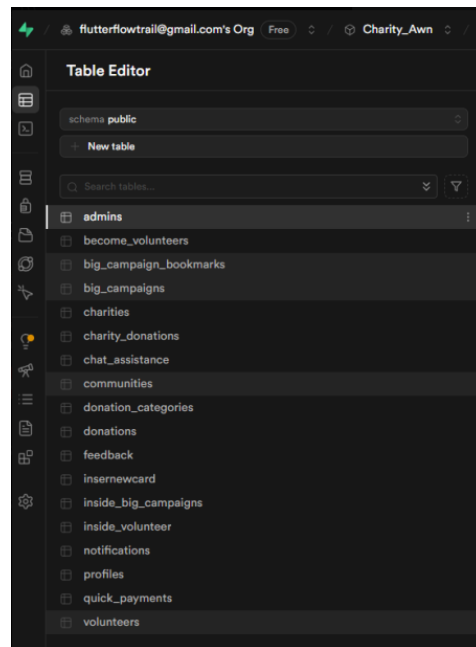


Figure 6.3.1- 1. Database Tables in Supabase

6.3.2 Authentication & User Roles

The Awn Charity App has Supabase authentication and user role management implemented to enable access control for users. The process of the authentication is to register with an email and password, login, logout, reset password, and verification of the OTP for validating a user account. It will validate the user at the time of activation of account and also when someone is recovering his/her account. The FlutterFlow emotion would allow Supabase authentication services to be integrated as an action. In this way, the emotion to log in securely can be done without backend programming.

There was a setup of user role to distinguish standard users and admins. The roles defined what the users are allowed to do in a system and what elements they can access. Standard users could donate, browse charities, volunteer, engage with communities, and manage their accounts. However, administrators were granted access to more system features. For instance, they were able

to manage the charity records. As well as the volunteer campaigns and donation categories. This setup for different roles secured the system and made sure that administrators were not tampered with.

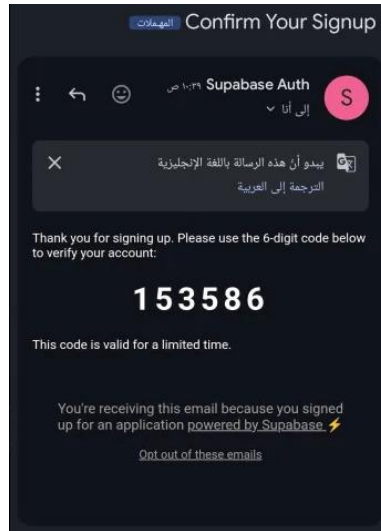


Figure 6.3.2- 1. OTP Email for Account Verification

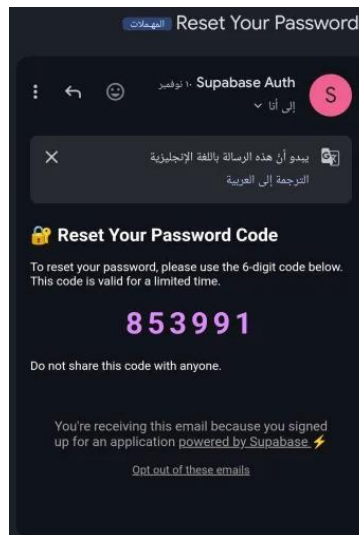


Figure 6.3.2- 2. OTP Email for Password Reset

6.3.3 Admin System Implementation

The Admin System allows the assigned administrators to access and manage crucial components of the Awn platform securely and centrally. The admin system covers a wide range of functionality including users, charities, volunteer opportunities, donations, and notifications. This system gives the administrators what they need to guarantee data accuracy, monitor activity, and support the system. This area is only accessible to people with higher privileges in order to ensure authorization and not misuse of controls.

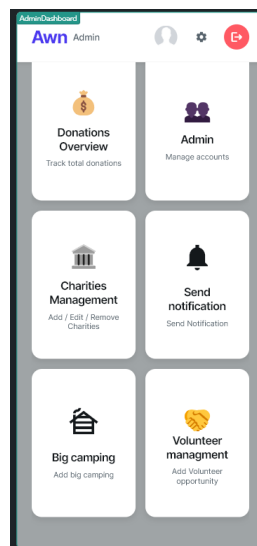


Figure 6.3.3- 1. Admin Dashboard Interface

As illustrated in Figure 6.3.3-1, the admin dashboard interface of the Awn is shown in the screen. It is the primary access point for all admin tools and controls. Administrators will have easy access to the key modules like the Donations Overview, Admin Management, Charities Management, Volunteer Management, and Notification System through this dashboard. The interface should be simple, organized and easy to navigate. Also, it should let the administrator perform system operation easily and clearly.

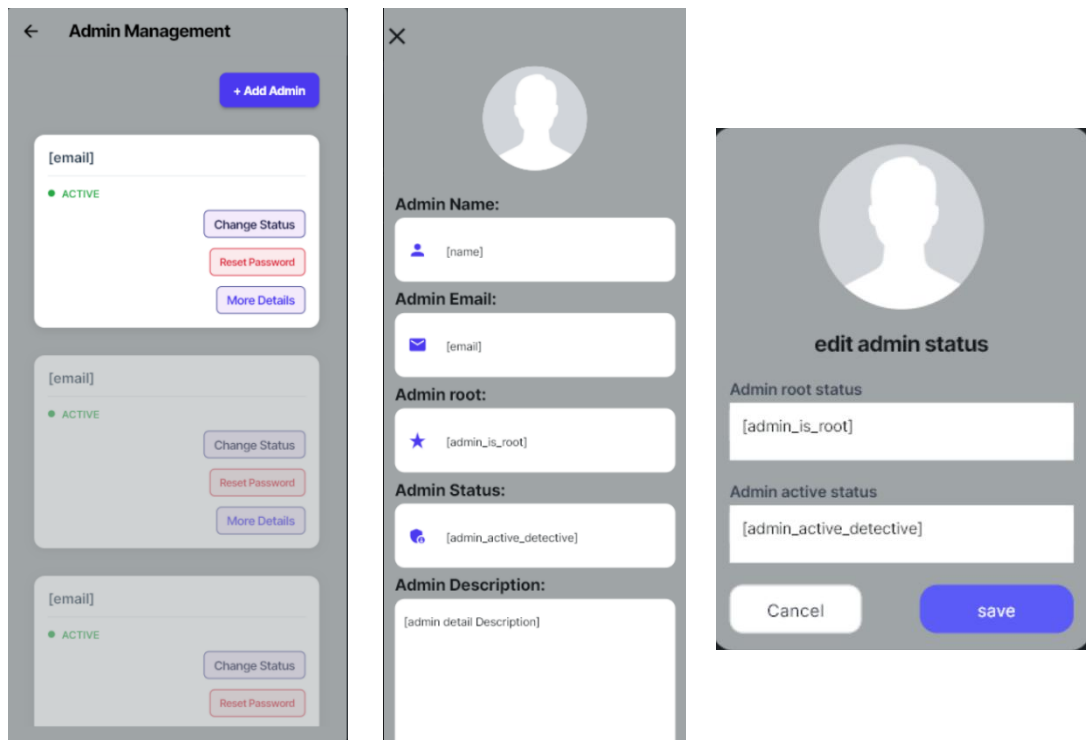


Figure 6.3.3- 2. Admin Accounts Management Screens

The interface depicted in Figure 6.3.3-2 is for the Admin Accounts Managements module of the AWN. Authorized administrators can create or edit new admin accounts or activate or deactivate them when necessary. While testing, we confirmed that all the actions taken on an account, like password updates and status changes, happened instantly on the database and user interface. Only those who are assigned the role of administrator, and are entitled to do so, can access this module.

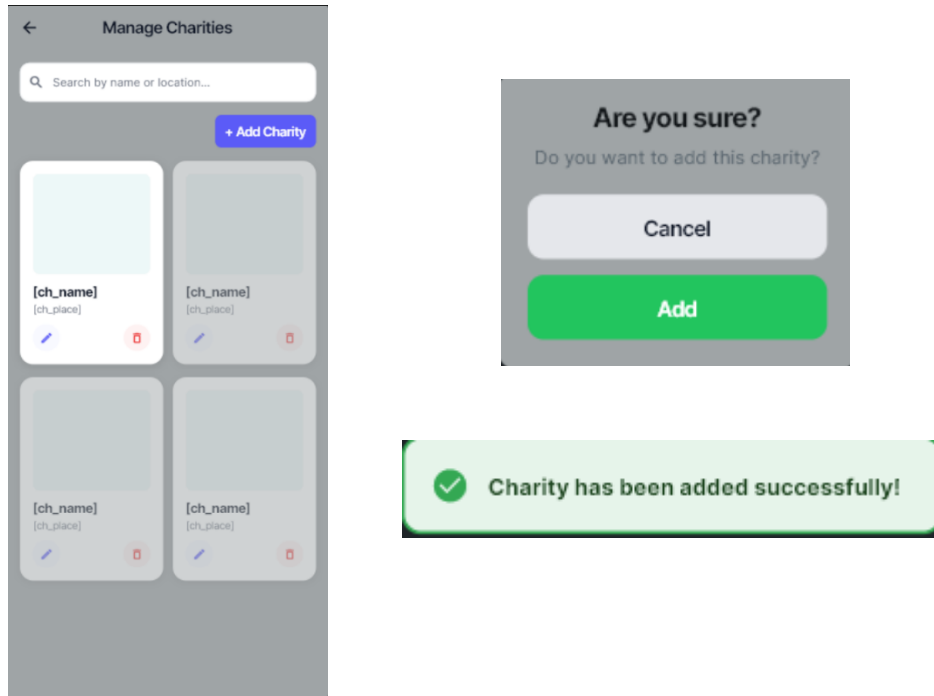


Figure 6.3.3- 3. Charity Management Screens

Figure 6.3.3-3 shows that through the Charity Management module, the administrator can effectively manage and maintain records of all charities registered on AWN. Using this interface, admins can add a new charity, update the existing one or delete the lesser used one. Before any critical operation, such as deletion or saving changes, a pop-up message will appear for confirmation. The updates for insert, edit, and delete were tested successfully and verified that everything synchronizes in the database to reflect on the platform. Overall, this module helps in keeping the charity information structured, updated and reliable for all the users of Bahrain Charity Network.

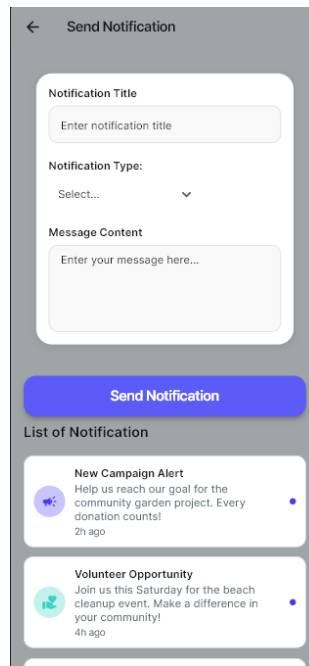


Figure 6.3.3- 4. Notification System Interface

As shown in Figure 6.3.3-4, the Notification System interface allows administrators to send messages to select users related to new campaigns, updates to the system, requests for volunteers, and other announcements. It is necessary to fill key fields for the effective communication of notification type, title, and message. Through testing, notifications were confirmed to show timely on the user interface in a structured and readable manner. There's also a notification history log for administrators to see what has already been sent and comment on it for auditing purposes.

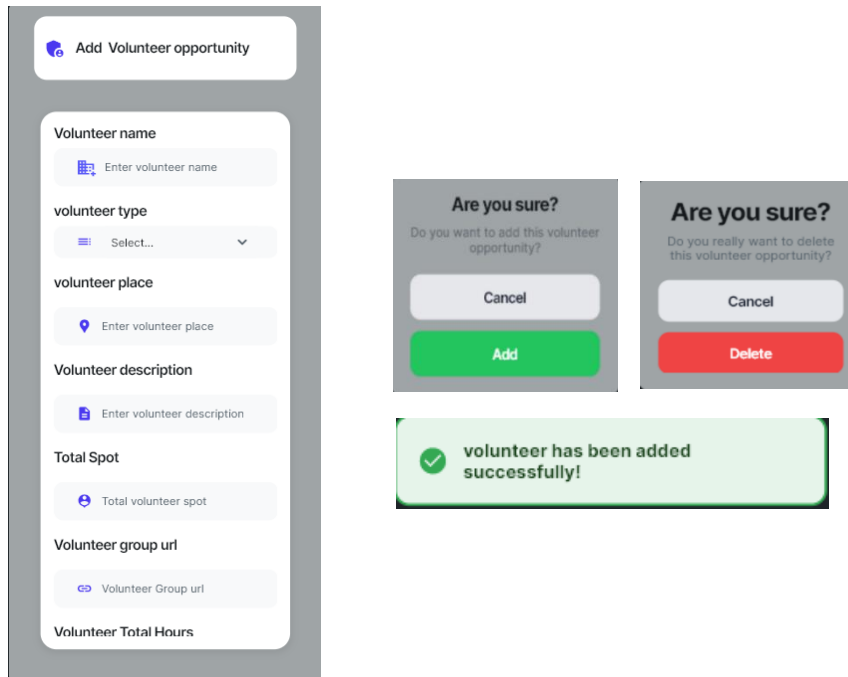


Figure 6.3.3- 5. Volunteer Management Screens

According to Figure 6.3.3-5, the Volunteer Management module allows an admin to manage all system volunteering opportunities. The admin can add a new volunteer opportunity using this interface by providing details like event name, category, location, number of slots and schedule. Ongoing opportunities can also be edited or updated at any time to ensure accuracy. Moreover, the deletion option can be used by the administrator to delete a possibility that had been inactive for long. Before any action is confirmed, similar to other admin components, a popup will appear to prevent confusion. The message on the page when success is confirmed is that the task is completed. These features help keep the volunteer listings accurate, neat and updated for all users on the platform.

6.3.4 Donation System Implementation (user)

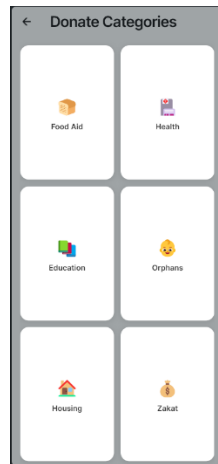


Figure 6.3.4- 1. Donation Categories Interface

As seen in figure 6.3.4-1, the Donation Categories Interface enables users to begin their donation process smoothly by selecting either 1 of the organized donation categories which include Health, Food Aid, Education, Zakat or Housing. Organizing donations allows the user to find out the cause of their liking. Moreover, it guarantees a clear and easy donation process. This structure makes decision-making simpler and encourages participation through an experience that is easy, intuitive and meaningful.

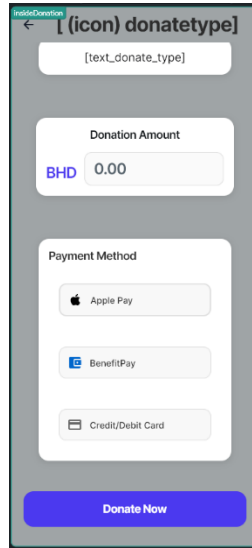


Figure 6.3.4- 2. Donation Amount and Payment Options Screen

Figure 6.3.4-2 shows the screen that appears after a donation category selection is made. Users can enter their donation amount and choose payment via Apple Pay, BenefitPay, or a credit/debit card stored with the platform. The design is basic and intelligible to users, allowing them to make a one-time payment or save the payment for later use. Supabase stores your payment and donation records after all the transactions have been securely processed.

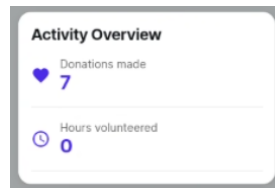


Figure 6.3.4- 3. Donation History Interface

As indicated in Figure 6.3.4-3 below, users can access all their previous donations through the Donation History. Every entry shows important details like the date when the donation is given, amount donated, and charity or campaign it supported. It enables users to view their donations, previous transactions, and to be transparent about their donations on the AWN Project.

6.3.5 Volunteer System Implementation (user)

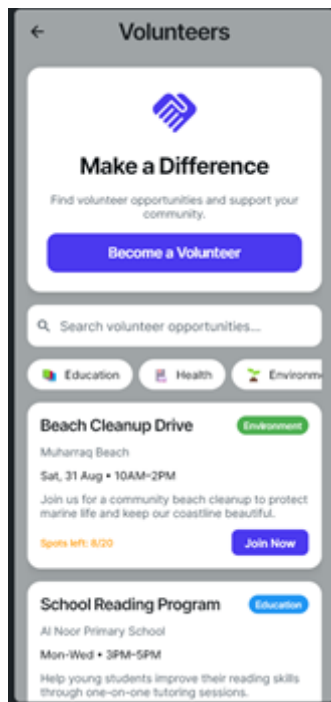


Figure 6.3.4- 4. Volunteer Opportunities Interface

As shown in Figure 6.3.4-4, the Volunteer Opportunities interface allows users to browse and join available volunteering events across the AWN platform. Each opportunity card includes key information such as the event name, category, location, schedule, and available spots. Users can search or filter opportunities to match their interests or availability. When selecting an opportunity, the user may join through the system, and if a communication link such as a WhatsApp group is provided by the administrator, tapping Join Now will redirect them to connect with organizers and other participants. This feature ensures smooth engagement and real-time coordination for volunteering activities.

The image displays three distinct mobile application screens. On the left is the 'Become Volunteering' form, which includes sections for 'Your Information' (with fields for Full Name, Email, Phone, Emergency Contact (Optional), and Notes) and 'Preferences & Consent' (with a Time Slot selector set to Morning, a Skills section with buttons for Driving, First Aid, Teaching, Translating, and Account, and a checkbox for 'I agree to event guidelines'). A blue 'Become Volunteer' button is at the bottom. To the right is a 'Confirmed Donation' screen featuring a green checkmark icon, the text 'Confirmed Donation' and 'Your Donation has been Sent successfully.', and a blue 'Done' button. Below these is a red error banner with a warning icon and the text 'please fill the fileds you are missed'.

Figure 6.3.4- 5. Implementation Become Volunteer

The Become Volunteer form, as illustrated in Figure 6.3.4-5, is used to register the user as a volunteer in AWN. This form collects necessary details including availability, skill sets and contact details. There are validation rules that ensure all the necessary fields are filled in. When you submit, a message appears. The information gets securely stored in the database. By registering on the platform users will be able to create a volunteer profile and track their engagement history.

6.4 Testing

The Awn Charity App testing section clarifies that the testing was done to ensure that all features work, and have a good experience, plus meet project requirements. Various testing techniques were used such as functional testing, usability testing, performance testing, and bug fixing. Our test procedure was to confirm whether the system is reliable and performs as expected. Next, we want to access issues, if any, and learn how stable the system is pre-deployed.

6.4.1 Functional Testing

System functional testing ensures that each functionality of the system operates in conformance with its requirement. We tested each core functionality of the platform individually, including authentication, donations, volunteer registration, admin controls, and notifications. Test scenarios were carried out real inputs and edge cases to ensure the system works way.

Table 6.4.1-1: Functional Testing Results

Test Case	Expected Result	Actual Result	Status
User login with correct credentials	User successfully logs in	expected	Pass
Incorrect login attempt	Error message displayed	expected	Pass
Donation submission	Donation saved and success message displayed	expected	Pass
Join volunteer opportunity	Redirects to WhatsApp (if link exists)	expected	Pass
Admin add charity	Charity added and visible in list	expected	Pass
Admin delete charity	Confirmation popup record removed	expected	Pass
Submit Become Volunteer form	Stored in database and success popup shown	expected	Pass

All core features passed functional testing, confirming that the expected behaviors were achieved.

6.4.2 Usability Testing

The usability test was to ensure that the application was easy to navigate, clear to read and intuitive to use. The system was evaluated at will and normal activities such as login, donating, exploring charities and volunteering were executed. As per the feedback from the Usability test, user completed the mentioned tasks without any confusion, while the interface had a clean look and was easy to read. Notifications, success messages and validation warning indicators clearing made the user aware of everything.

6.4.3 Performance Testing

The speed and stability of the system while performing various operations were assessed through performance tests. The app was tested under a variety of network conditions. Results showed:

- Fast loading of screens
- Smooth transitions without delays
- Stable database syncing
- Make sure to handle when the user repeats or acts quickly

The system functioned correctly and efficiently, even after switching modules or sending large number of requests.

6.4.4 Bug Fixing

While testing, a few errors popped up regarding the form validation, where the new control should be located, and the error messages that were not coming in. Each of these issues has been documented, corrected, and retested. After adjustments, everything worked as expected; all modules function correctly with no major error.

6.5 Summary

The chapter discussed the development and validation of the Awn Charity App through Implementation and Testing. The implementation phase showcases the development and integration of core system features, namely authentication, donations, volunteering, and admin, via FlutterFlow and Supabase. The way the system was built and how it runs to help the users and the administrators was shown through screenshots, workflows, and module descriptions.

The system was found to work efficiently, have a user-friendly interface, and perform reliably. We validated the functionality with functional testing, ensured the interface is user-friendly with usability testing, checked responsiveness under various conditions with performance testing, and bug fix issues. With this, the results indicate that the system is stable, functional, and fit for deployment and for use in real life.

Chapter 7

Conclusion and Future Work

7.1 Introduction

The project involved mobile application development where the team gained experience in the design, development, and management of a real-life application. We used our technical and analytical skills to respond to a real problem that brought together all charities in Bahrain onto one easily accessible platform while developing the AWN Charity App. We understand making experience-centered human making. Making cloud-based systems (FlutterFlow, Supabase) real-time and integration (for system stability, scalability). The project also helped us improve as a team and communicate better. We have to faced challenges like syncing charity data, integrating payments, usability for different users (donor, volunteer, admin) and more. By overcoming these challenges, I learned best practices in software engineering as well as real-life digital transformation issues in charities.

7.2 Conclusion

After completing the AWN Charity App project, we have built a platform that connects donors and volunteers with NGOs across Bahrain. We faced multiple challenges during the process such as merging multiple databases of different charities, ensuring online payment with BenefitPay and card payment was secure, and the design was user-friendly for all users. This experience has developed technical, analytical and team working skills to great extent. Using FlutterFlow, Supabase, and real-time data synchronization, we learned about mobile application development. Also, I learned that testing and feedback, project planning, and work are essential for making a credible social-impact app. Our project strengthened sense of community responsibility. Therefore, we made our contribution toward Bahrain's charity ecosystem's digital transformation. The AWN application is an important step to simplify and modernize charity communications in the Kingdom, though there is still more to do.

7.3 Future Work

AWN future vision is to improve the platform further and include additional features, as well as use beyond the application. We want AWN to be a complete digital ecosystem experiencing that consists of a mobile app and responsive website enabling users to access all of our charitable services from any device. The website will have the same experience as the mobile one but will also give more admin and data tools to charity organizations. We will also develop a dashboard to track donor progress as well as volunteer performance tracking that personalizes engagement and optimizes user satisfaction through AI-backed campaign recommendations.

Furthermore, we also plan to introduce multi-language support to make AWN available to Bahrainis and expatriates alike. An objective of AWN is to be the number-one charity platform in Bahrain, essentially serving as the official charity platform for all charity events in the Kingdom. AWN wishes to launch and manage the major fund-raising campaigns, so that they are completed in record time and transparency. Through automation, secure payments, and real-time analytics, it will make it easier for donors, volunteers, and charities to work together. AWN aims to expand regionally in the long run along with being the general model for consolidated charitable systems in the Gulf. AWN will transform how people contribute, volunteer, and engage for community benefit through a smarter, reliable and inclusive platform. It further strengthens its commitment by collaborating with government institutions, NGOs and fintech companies.

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